FRC 2025 REEFSCAPE -- Robot Design Deep Dive

Design reference for REEFSCAPE. Each profile comes from the team's own public material: CAD (Onshape, GrabCAD, Fusion 360, STEP), technical binders, code, and Chief Delphi release and reveal threads. Performance data comes from The Blue Alliance.

The REEFSCAPE design landscape

The game

REEFSCAPE is a coral and algae game. Robots place CORAL (PVC tubes) on the four branch levels of the REEF (L1-L4). Robots score ALGAE (foam balls) in the PROCESSOR or the BARGE NET. In the endgame, robots park or climb a CAGE (shallow or deep). Many designs used an elevator or an arm to reach L4. Some top teams scored coral only and did not handle algae.

How to use this document

The first section lists the profiled robots in the top 30 by Local EPA. The second section lists the other profiled robots. Each profile gives the archetype, the subsystems, vision, endgame, autonomous routines, the public CAD or binder, and lessons for students. A star marks a study pick. Each study pick has downloadable CAD or a full technical binder and one specific design lesson. "Unconfirmed" means the public material does not state the detail. It does not mean the robot does not have it.

About the EPA numbers

The Statbotics data API is offline. EPA is Local EPA (approx): it was computed from The Blue Alliance match data with the open-source Statbotics model (github.com/avgupta456/statbotics). The rank order follows Statbotics. The absolute values can differ from statbotics.io. Records are win-loss-tie totals from official events on The Blue Alliance.

Profiled robots at a glance (sorted by Local EPA)

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**2056 OP Robotics**	LIGHTNING	**123**	#1	68-7	'LIGHTNING': swerve; counter-rotating roller coral intake; riveted she	Tech binder
■	**2910 Jack in the Bot**	Poi Pounder XXVI Spectre	**113**	#3	66-7	'Spectre': shoulder, extension and wrist arm; latch climber with autom	Onshape CAD
■	**1678 Citrus Circuits**	SubLime	**109**	#4	57-8	'SubLime': sideways elevator on Dyneema cable at 3:1; coral and algae	Onshape CAD
■	**1323 MadTown Robotics**		**106**	#5	53-3	Ground coral and algae intake with auto handoff; elevator on a pivot;	Drive folder / doc
■	**422 The Mech Tech Dragons**	Wisp	**95**	#14	60-8	'Wisp': coral ground intake (pivot began as algae roller), polycarbona	Onshape CAD
■	**1778 Chill Out**	SubZero	**93**	#17	53-12	'SubZero': ground coral intake with auto handoff to a 580 deg arm on a	Onshape CAD
■	**604 Quixilver**	The Flush	**92**	#18	39-10	'The Flush': WCP X2i swerve; single-stage elevator, arm and pivoting o	Onshape CAD
■	**3005 RoboChargers**	Relay	**91**	#19	56-9-2	'Relay': 3-stage chain-driven elevator carrying a 'laterator' with a c	Onshape CAD
	180 S.P.A.M.	Anemone	**90**	#23	44-6	'Anemone': floor-pickup coral robot run by one driver, with automated	Code (GitHub)
■	**5990 TRIGON**	TRIDOG	**90**	#22	46-13-2	'TRIDOG': ground coral intake (about 7 kg); elevator on custom-machine	Onshape CAD
■	**7407 Wired Boars**	Griptide	**90**	#24	55-13	'Griptide': coral robot, coral station intake, 3-stage elevator, rotat	Onshape CAD
■	**254 The Cheesy Poofs**	UNDERTOW	**89**	#26	42-7	'UNDERTOW': coral and algae robot; 4-bar main intake, backup coral sta	Tech binder
■	**190 Gompei and the HERD**	StackUp (V1) and Redundancy (V2)	**88**	#30	68-19	Two released robots (V1 StackUp, V2 Redundancy): 4-motor elevator, sta	CD thread
■	**341 Miss Daisy**	Miss Daisy XXIII	**88**	#28	54-12	Miss Daisy XXIII: coral floor intake, continuous elevator, lantern-gea	Onshape CAD
■	**9450 Velocity Raptors**		**88**	#27	50-17	Elevator with a differential wrist under custom control code; 3 piece	Onshape CAD
■	**1732 Hilltopper Robotics**	Private	**86**	#34	41-17	'Private': 4x L4 coral auto with QuestNav localization; algae handoff;	Onshape CAD
■	**1114 Simbotics**	Simbot Suzuki	**83**	#52	43-22	Station-fed coral elevator with reef algae removal; swerve; deep cage	Onshape CAD
■	**581 Blazing Bulldogs**	Zuma	**82**	#59	33-16	'Zuma' (Champs): 2-stage elevator, slip-ring arm with unlimited rotati	Onshape CAD
■	**6800 Valor**	Deadlock	**82**	#57	59-10	'Deadlock': 3-stage elevator; funnel; 3 non-drive motors pre-Worlds, 5	Onshape CAD
■	**9496 LYNK**	Chimney Rock (Rocky)	**81**	#69	72-7-2	'Chimney Rock' (Rocky): low-rider chassis on SDS L2 swerve; passive ga	Onshape CAD
■	**971 Spartan Robotics**	Fiddler	**80**	#84	30-18	'Fiddler': swerve after tank in 2023-2024; stiffened elevator with arm	Onshape CAD
	2630 Thunderbolts	Atlas	**80**	#86	47-16-1	'Atlas': L4 coral scorer with algae into the net; 4 L4 coral auto in 1	Onshape CAD
■	**3467 Windham Windup**		**79**	#89	49-19-1	Elevator with coral claw; algae in horns or a lower claw; deep cage cl	Onshape CAD
■	**2930 Sonic Squirrels**	Holocentrus	**78**	#100	46-22	'Holocentrus': elevator robot; 100+ short Choreo paths; PhotonVision a	Onshape CAD
■	**695 Bison Robotics**	Goldfish	**76**	#124	26-14	'Goldfish': elevator coral scorer, ramp-fed pivoting arm, modified Eve	Onshape CAD
■	**3128 The Aluminum Narwhals**	Gillbert	**76**	#122	31-13	'Gillbert': elevator robot; sensorless counter-rotating coral end effe	Onshape CAD
■	**5460 Strike Zone**	TITAN	**75**	#141	52-12-2	'TITAN': continuous elevator with a 50 mm shaft arm; rotating coral cl	Onshape CAD
■	**294 Beach Cities Robotics**	Sacabambaspis	**73**	#160	27-18	'Sacabambaspis': station hopper; pivoting wrist with pass-through side	Onshape CAD

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**6995 NOMAD**	Mariana	**73**	#163	54-16	'Mariana': elevator and pivot with a two-sided ground 'hand' for coral	Onshape CAD
■	**3061 Huskie Robotics**	Calypso	**70**	#220	25-15	'Calypso': SDS MK4n swerve; side-to-side funnel into a four-stage elev	Onshape CAD
■	**6328 Mechanical Advantage**		**70**	#209	44-21	Dev bot plus four competition CAD versions (25B, 25C, 25D, 25O); arm p	CD thread
	3414 Hackbots	Levi	**69**	#233	34-26	'Levi': elevator with a fixed-angle coral end effector and a separate	CD thread
	5419 Berkelium	Argonaut	**69**	#227	27-17	'Argonaut': coral-focused robot with elevator, pivot and climb; auto-a	Onshape CAD
■	**1156 Under Control**	Ultralisk	**68**	#243	32-12	'Ultralisk': swerve with an always-on heading PID set by the right-sti	Onshape CAD
■	**8159 Golden Horn**	Serpentheim	**66**	#295	32-11	'Serpentheim': coral funnel and a separate algae system; two cameras;	Onshape CAD
■	**4795 Eastbots**		**64**	#314	41-19	Elevator and chain-driven arm with a motorized wrist (added for DCMP);	Onshape CAD
	418 Purple Haze		**62**	#362	36-24	Sideways 1-stage elevator + arm ('Armavator'); 'Thagomizer' coral scor	Onshape CAD
	1153 Timberwolves	Lagotto	**62**	#369	46-24	'Lagotto': 1-stage elevator scoring coral L1-L3 and algae in processor	Onshape CAD
■	**587 The Hedgehogs**		**61**	#372	38-20-2	Hopper-fed coral scorer on an elevator with a pass-through wrist; full	Onshape CAD
	233 The Pink Team	Bubbly Anemone Enemy (BAE)	**58**	#458	36-17-1	'Bubbly Anemone Enemy': chain-and-Dyneema cascade elevator, coral shoo	Onshape CAD
	1710 The Ravonics Revolution	Dobson	**57**	#483	25-13	'Dobson': elevator robot for coral L1-L4; algae claw for reef algae an	Fusion 360 CAD
■	**3357 COMETS**	Tempel	**56**	#490	23-21	'Tempel': 24x24 in frame, ground coral, sideways belt-in-tube elevator	GrabCAD
	2485 Overclocked	Nessie	**53**	#581	20-16	'Nessie': elevator robot with a bent polycarbonate coral ramp into the	Onshape CAD
■	**3669 CarbonKnights**	Twitchy	**52**	#593	9-13-1	'Twitchy': two parallel carbon fiber tube elevators with 3D printed be	Onshape CAD
	1086 Blue Cheese	Stringray	**50**	#657	31-25-1	'Stringray': coral station intake, coral L1-L4, algae from reef to net	Onshape CAD
	1740 Cyber Colonels		**42**	#988	12-17-1	January plan: coral-only RustHOUNDS Ri3D-style robot, passive ramp int	Onshape CAD
■	**614 Night Hawks**	Stand Mixer	**41**	#1010	19-24-1	'Stand Mixer': 2-in-1 ground coral intake (L1 hold or pass-through) fe	Onshape CAD
	7312 T3	Nessie	**35**	#1349	24-18	'Nessie': 3-stage GreyT cascade elevator; elbow and wrist pivots; belt	Onshape CAD
	2079 4H ALARM Robotics		**29**	#1774	9-19-1	28x28 swerve, 2-stage cascade elevator, wrist carriage with top-down r	Onshape CAD
	7691 TSA All Stars		**25**	#2124	18-24-1	Swerve; elevator with arm end effector for coral; pivoting ground cora	Onshape CAD
■	**1806 S.W.A.T.**	Atom	**10**	#3554	9-16	'Atom': swerve, custom welded elevator; rebuilt mid-season toward a 20	Onshape CAD

■ = study pick: downloadable CAD or a full technical binder, and one specific design lesson. EPA is Local EPA (approx), computed from The Blue Alliance match data. Records are win-loss-tie totals from official events.

Top-30 robots by Local EPA

■ Study pick

2056 OP Robotics LIGHTNING

Canada

Local EPA 123 (#1)

68-7

'LIGHTNING': swerve; counter-rotating roller coral intake; riveted sheet-metal arm; 2D AprilTag auto-align

SCORING (CORAL / ALGAE)
Coral and algae end effector unconfirmed (see the drawing package). The team's posts describe a 'dunk' scoring step that the driver triggers after auto-align. The driver keeps this step so the robot can rescore after a miss and lose a second or two, not a full cycle.

INTAKE
Roller intake with top rollers and a bottom roller that turn in opposite directions; a small 1:1 gearbox reverses the bottom roller. The front roller floats, and coral compression changes with coral orientation. WCP stretch star wheels sit on 1.5 in OD, 1/8 in wall polycarbonate tube. Changed after practice from 1.25 in OD, 1/16 in wall tube after the tube bent in practice and the wheels migrated and slipped. The polycarbonate pivot is clamped with aluminum plates that spread bolt loads in impacts.

ELEVATOR / ARM
The arm is a riveted and formed 1/16 in aluminum structure. The robot uses 5052 aluminum sheet at 1/16 in and 1/8 in depending on load. The team states it drives pivots with large aluminum sprockets on a dead axle and chain: #25 for light-duty and #35 for heavy-duty pivots. Elevator details unconfirmed (see the drawing package).

DRIVETRAIN
Swerve. The team states its swerve control and odometry are fully custom, not CTRE's generated swerve code. They use Phoenix Pro features: FOC, fused CANcoder and time synchronization. Module type unconfirmed (see the drawing package).

ENDGAME
Unconfirmed (not stated in the public release).

AUTONOMOUS
Waypoints are hard-coded points, with no path-planning interface. The tolerance per waypoint ranges from 1.5 in to 48 in, depending on the action. In auto, the robot computes the coral position and generates a drive point with a heading that points at the coral. Auto selection inputs are read only after a 3 s timer in disabled, so the driver cannot change them between auto and teleop. Coral count per routine unconfirmed.

VISION
Limelight with 2D AprilTag tracking only; the team does not use MegaTag 1 or MegaTag 2. Distance comes from an interpolating tree map of tag area to measured distance. Auto-align drives to a pose relative to the reef tag for L2-L4 and uses global pose for L1. A drive-to-point controller runs two PID loops, on translation and rotation error, with an aggressive slew rate limiter.

PUBLIC CAD / DESIGN
Technical binder (PDF)

LESSONS FOR STUDENTS

- Drawings can document a robot without a CAD model: the drawing package keeps every part revision, marked R1, R2 and so on, so a student can follow the changes through the season.
- Auto-align without a field pose: the robot measures the reef tag relative to itself and drives to a pose a set distance and heading from it, using global pose only for L1.
- A material fix after a practice failure: the star-wheel roller tube went from 1.25 in OD, 1/16 in wall to 1.5 in OD, 1/8 in wall polycarbonate after it bent and the wheels slipped.
- Aluminum plates on a polycarbonate pivot act as large washers and spread bolt loads, so side impacts do not crack the polycarbonate at the bolt holes.
- The team prefers chain to 5M HTD belt on pivots because chain is less likely to skip under shock loads, and designs in a chain tensioner, such as the inline turnbuckle tensioner used on the intake.

Tech binder: Technical binder (PDF) · Binder release thread (CD) · Technical binder (PDF) · CAD drawing package (PDF) · Auto-align approach post (CD) · Intake slow-motion video (Google Photos)

■ Study pick

2910 Jack in the Bot Poi Pounder XXVI Spectre

USA-WA

Local EPA 113 (#3)

66-7

'Spectre': shoulder, extension and wrist arm; latch climber with automated climb; reef-AprilTag auto-align; swerve

SCORING (CORAL / ALGAE) End effector on the arm wrist. It holds coral straight, as the intake delivers it, or horizontal in an L1-optimized position, and it also handles algae. Scoring uses one button for a 'base' setpoint and one for a 'top' setpoint. Translation, rotation, release, wait and backout are all automated. Rollers and motors: Unconfirmed (see CAD and tech binder).	**INTAKE** Intake type: Unconfirmed (see CAD and tech binder). The team states it never used coral station intaking during teleop. On station loading, a team reply says it is better to drive firmly into the station and guarantee the load than to risk wheel slip leaving the robot short.
ELEVATOR / ARM Arm with a shoulder pivot, an extension and a wrist. The code treats each as its own subsystem, which the team says gives cleaner compartmentalization. Shoulder and wrist are commanded by angle. Positions come from the built-in Kraken encoders on the shoulder, extension and wrist. A re-homing button was used at most twice in the season, after heavy arm impacts; it mainly saves time in driver practice. The arm code ran on the team's 2023 robot before bring-up. Motor count unconfirmed (see CAD).	**DRIVETRAIN** Swerve; the team describes hard-resetting the swerve odometry from vision. Module type unconfirmed (see CAD). A skew compensation constant corrects drift off center when the robot rotates while translating. The code uses voltage targets, not duty cycle, so output holds as battery voltage drops; the reply does not say which motors.
ENDGAME Climber with latches and climber intake wheels. The climb is fully automated once the bar is in the latches. The code detects the bar from the speed and current of the climber intake wheel motor, and the team reports no false positives.	**AUTONOMOUS** Autonomous Award at PNW District Auburn, Sammamish and the PNW DCMP. The team mentions a 'mark auto' and tested code to knock algae off the mark; that code was not used. Routines and piece counts unconfirmed (see code).
VISION Cameras read reef AprilTags. There is no pose estimator: the code hard-resets swerve odometry to camera poses, and only while auto-aligning, when the robot is stable. The team reports vision as accurate at the scoring position, with more error farther out. Drivers switch between pose-based and rotation-based reef selection; LEDs show purple for pose-based and yellow for rotation-based.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A plain PID auto-align with a velocity cap and friction compensation can be enough: the team tried trapezoidal profiles and dynamic path following and kept PID.
- Relocalize only when the data is good: odometry resets from vision happen during auto-align, not during fast traversals.
- Give drivers a fallback when odometry drifts: rotation-based reef selection starts auto-align without trusting pose, and LEDs show the mode.
- Motor data can replace a sensor: the climber detects the bar in the latches from intake-wheel speed and current.
- Bind buttons to robot state: one button does four things depending on the piece held (straight coral, horizontal coral, algae, none). Drivers use Xbox controllers with back paddles. The arm code was proven on the 2023 robot before bring-up.

Onshape CAD: Onshape CAD · CAD and tech binder release (CD) · Onshape CAD · Tech binder (Google Drive) · Code (GitHub) · Code release thread (CD) · Robot reveal (CD) · Reveal video

■ Study pick

1678 Citrus Circuits SubLime

USA-CA

Local EPA 109 (#4)

57-8

'SubLime': sideways elevator on Dyneema cable at 3:1; coral and algae end effectors; indexer; swerve

SCORING (CORAL / ALGAE) The team's 2025 SystemCore thread lists separate beam breaks for a coral end effector and an algae end effector. All scoring runs on one controller through back paddles, with an autoscore function. End effector rollers, motors and geometry: Unconfirmed (see CAD).	**INTAKE** The SystemCore thread lists an indexer beam break. Coral station or ground intake and algae intake: Unconfirmed (see CAD).
ELEVATOR / ARM Sideways elevator driven by Dyneema cable at a 3:1 reduction; the OP gave the ratio in the reveal thread, correcting a reader's 4:1 estimate. The team has used the same cable (CBKnot 7/64 in Dyneema, about 1600 lb tensile strength) on all its string elevators since 2018. The elevator support plate and the sponsor panel protect the cable. It stretched slightly and was re-tensioned when loose; the team replaced it once during the season. The sideways layout let the stages rock front to back with the elevator up, which hurt L4 accuracy. The design lead measured the gap and cut new stage 0 top spreader plates with offset holes. Most of the loose-fit slop went away; the middle-stage tubes still deflect under drive acceleration. Elevator motor count: Unconfirmed (see CAD).	**DRIVETRAIN** Swerve, per the SystemCore thread. That thread traces a fault on the back left module of the practice robot 'Omega' to a faulty MPM port for its CANcoder. The robot runs two CAN loops, one on the roboRIO and one on a CANivore. Module type: Unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** The reveal thread links a video of an auto routine in development (2025-02-25). Coral count and path tool: Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Measure slop before you redesign: the team measured the front-back gap in its sideways elevator and fixed most of it with new spreader plates that have offset holes.
- Cable-driven elevator: the team uses Dyneema cable as a lighter option for powering the elevator. Plan for re-tensioning, because the cable stretches slightly over time.
- Bent parts: the team makes all bent aluminum parts from 5052 sheet, which it describes as more ductile than 6061, and reports no cracking.
- Sensor layout: beam breaks on the coral end effector, the algae end effector and the indexer tell the code where each game piece is.
- Two near-identical robots: the team 'leap-frogged' updates, so one robot was upgraded while the other ran driver practice and software. The practice robot 'Omega' later tested a new controller.

Onshape CAD: Onshape CAD · Onshape CAD · CAD, code and strategy release thread (CD) · CAD and code release page · Robot reveal (CD) · SystemCore test thread (CD) · Auto in development (video)

■ Study pick

1323 MadTown Robotics

USA-CA

Local EPA 106 (#5)

53-3

Ground coral and algae intake with auto handoff; elevator on a pivot; differential wrist (pitch, roll); swerve; 3x Limelight 4

SCORING (CORAL / ALGAE)
A claw on a differential wrist scores coral and algae. The team runs the wrist in CTRE differential mode and lists its hurdles: fitting parts in the carriage, belt slip, and enough encoder wrap for a 1:1 reading. It used standard bevel gears, not spiral, and says it would try herringbone-style bevels next time. A retroreflective Banner sensor on a CANdi in the claw detects both pieces. The end effector motor runs in PWM mode to avoid a long CAN stub, after stator and current limits were set over CAN. A forum question names the motor as an X44; the team's answer does not restate the model.

INTAKE
Ground intake for coral and for algae. A diffuse Banner sensor on a CANdi sees the coral; the robot then lifts, hands off and enters a scoring state on its own. Ground algae hands off from the ground intake to the claw the same way. Reef algae pickup runs from a driver trigger: the robot squares up and follows a path to the ball, which lets the co-driver focus on lowering the coral intake.

ELEVATOR / ARM
The code models each position as a SuperstructureState with four values: elevator, elevator pivot, wrist pitch and wrist roll. A 'choreographer' compares current and target positions and builds a command that moves between them safely. Elevator tube is mainly .060 in wall for weight; each stage is about 21.25 in tall. Stage count: Unconfirmed (see the tech binder). Thinner 12/14 AWG wire and Alpha Wire let the team drop from a 15 x 30 mm to a 15 x 15 mm e-chain.

DRIVETRAIN
Swerve, frame perimeter about 112 in per the team. A forum user read WCP X2 modules in two configurations from the tech binder; a team member stated the mix was for space and packaging. The A-frame tube support uses a 3/16 in aluminum plate with custom nut plates inside the 2x2 in box tube, plus a .090 in bent aluminum plate. The team uses the top of the swerve plate as structure, an idea it credits to 4414.

ENDGAME
Unconfirmed (not stated in the public release).

AUTONOMOUS
Autonomous Award at the Silicon Valley Regional. Autos start from a pre-defined path; Limelight object tracking builds a new path to the coral, which the team says happens 99.9% of the time. The camera tracks the ball and checks its height off the ground to confirm the coral is still there. The team has used the built-in Limelight model since 2023. Coral count: Unconfirmed (not stated in the public release).

VISION
Three Limelight 4 cameras. Two read AprilTags without the Hailo accelerator; the team saw better performance without it. The third runs object detection in auto only. Every camera has a milled heat sink; two of the three have fans.

PUBLIC CAD / DESIGN
2025 Tech Binder (Google Drive)

LESSONS FOR STUDENTS

- Plan the weight budget early: the practice robot weighed 130 lbs in week 5, and the team lists each step it used to cut weight, down to 0.01-0.05 lb at a time.
- Run the practice robot hard and print its parts in a weaker material: PETG HF parts fail earlier than the Markforged Onyx parts on the competition robot, and the competition robot sees only a few auto tunes and matches.
- Wire Banner sensors through a CANdi: the team states detection was faster than with sensors plugged into the roboRIO.
- Name superstructure positions as states: each state sets elevator, elevator pivot, wrist pitch and wrist roll, and one 'choreographer' plans the safe move between states.
- Use object tracking to correct auto paths: the team says it would not grab the coral without Limelight tracking that rebuilds the path.

Drive folder / doc: 2025 Tech Binder (Google Drive) · 2025 Tech Binder (Google Drive) · Tech binder release thread (CD) · Reveal thread (CD)

■ Study pick

422 The Mech Tech Dragons Wisp

USA-VA

Local EPA 95 (#14)

60-8

'Wisp': coral ground intake (pivot began as algae roller), polycarbonate coral-retention flap, bellypan-keyed climber

SCORING (CORAL / ALGAE) A 1/16 in polycarbonate flap retains coral. The team balanced its stiffness so it holds coral at low current; at higher motor current the coral pushes through. The post does not name the mechanism that carries the flap. End effector type and roller motors unconfirmed (see CAD and tech binder).	**INTAKE** Ground coral intake on a pivot. At the first few events the pivot ran as an algae over-the-bumper roller; the team then swapped to the coral ground intake. SplineXL OD hubs were machined down for a press fit on SplineXL extrusions. After the swap the team drilled and tapped a set screw into each hub.
ELEVATOR / ARM Elevator / arm unconfirmed (see CAD and tech binder).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The battery is held by a nylon strap with a zip tie as the buckle; a billet holder was left off to save weight.
ENDGAME Climber fastened to the bellypan: a 2x2 tube plug inside the climber takes the bellypan bolts, and the gearbox plate profiles key into the bellypan. Deep or shallow climb unconfirmed (see CAD).	**AUTONOMOUS** Routines and piece counts unconfirmed (see tech binder).
VISION Pose estimation; the team does not say it is vision-based. It did not work on day 1 of the first event, despite working on a local practice field, and the team reworked it during that event. Cameras unconfirmed (see tech binder).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A tuned polycarbonate flap can retain a game piece with no extra actuator: it holds coral at low current and releases at higher current.
- One pivot can change jobs mid-season: the algae over-the-bumper roller became the coral ground intake.
- A press fit that holds for one load case may need a set screw for the next: the team added one when the pivot became the coral intake.
- Test pose estimation at an event, not only at home: it failed on day 1 of the first event and was reworked there.
- Battery condition matters late in an event: the team reports brownouts 10+ times per match in DCMP playoffs after its batteries degraded.

Onshape CAD: Onshape CAD · CAD and tech binder release (CD) · Onshape CAD · Robot reveal (CD)

■ Study pick

1778 Chill Out SubZero

USA-WA

Local EPA 93 (#17)

53-12

'SubZero': ground coral intake with auto handoff to a 580 deg arm on a belt elevator; pose-based auto-align; no climber

SCORING (CORAL / ALGAE) End effector (manipulator) on the arm, with a roller reduction of about 11:1. Current sensing detects coral; the team tuned a voltage that stalls the roller and still grips the coral. To score, the arm moves down with the elevator while the bumpers rest against the reef. Scoring is semi-automatic: when the driver releases the sticks, the robot drives to the nearest scoring position and picks the closer branch. The driver then presses a confirmation button to start the scoring sequence.	**INTAKE** Ground coral intake. Wheels inside the intake center the coral, and the upper wheels spin it as it centers. One linebreak in the center detects coral; the intake then raises and hands off automatically. Two X60 motors drive it; the team says that is a lot of weight and plans X44s in future. The bottom roller has bearings but the team stopped it from spinning, which it says improves handoff and trough scoring. The intake zeros by current sensing.
ELEVATOR / ARM Arm on a belt-driven elevator. The arm pivot has a 64.5:1 reduction. Software limits the arm to 290 deg left or right of straight down (580 deg range) and clamps the encoder range to 700 deg for safety. A WCP CAN throughbore encoder on a 2.22:1 reduction gives 800 deg of range and sets the arm position at match start; there are no hard stops. Arm wires pass through the pivot tube in wire wrap with slack. The elevator runs two open-end PolyBelt belts of about 13.5 ft, clamped to the carriage between aluminum and a profiled PLA part. Two motors with 9-tooth pinions drive 36-tooth gears, each on a shaft with a 24-tooth pulley. The carriage weighs about 7 lb, about 13.5 lb with the arm. The elevator zeros by current sensing. The team reports some elevator wobble and says the A-frame should have run to the corners.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME No climber. The team states the robot was at max weight, a climber would have needed many parts lightened, and it had about a week to build an untested climber. It used the time for LEDs and improvements to scoring, autos and the state machine.	**AUTONOMOUS** The auto video shows the ground centering and handoff sequence. Coral count unconfirmed (see code).
VISION Full-field pose estimation drives reef alignment. The team chose it over AprilTag tx/ty because the scoring point is offset from the robot center. Camera type and count unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Pick an absolute encoder ratio above 1 and not a whole number when an arm turns more than one revolution: 2.22:1 gives 800 deg of encoder range for a 580 deg arm.
- One centered linebreak can trigger intake raise and handoff if the coral is already centered by the time the intake is up.
- Locking a roller can help: the team stopped the bottom intake roller from spinning to improve handoff and trough scoring.
- Use full-field pose, not tag tx/ty, when the scoring point is offset from the robot center.
- Decide late additions by weight and time: at max weight with little room, the team skipped a climber and improved scoring and autos instead.

Onshape CAD: Onshape CAD · Onshape CAD · CAD and code release thread (CD) · Code (GitHub) · Reveal thread (CD) · Reveal video · Auto handoff video

■ Study pick

604 Quixilver The Flush

USA-CA

Local EPA 92 (#18)

39-10

'The Flush': WCP X2i swerve; single-stage elevator, arm and pivoting open-top gripper; full-width coral funnel; winch climb

SCORING (CORAL / ALGAE)
Gripper pivots 180 deg on the arm, with 4 coral rollers and 2 algae rollers. Left and right sides run independently to turn coral sideways for L1; it scores up to 4 coral per side on L1. The open top lets coral rotate on exit for a shallower L4 angle and keeps the robot from catching on the reef. Wrist gearbox is shimmed to decrease backlash. Gripper motors unconfirmed (see CAD).

INTAKE
Full-width funnel of 4 mm corrugated plastic for loading from 3 sides of the robot. Two separately powered rollers (top and bottom) work with the gripper's left and right rollers to center coral in any orientation. A rear ramp unfolds to clear the climber and cage. No ground intake: a summer 2025 ground intake for The Flush went on the offseason robot instead, because it was hard to package with the climb.

ELEVATOR / ARM
Single-stage elevator with 2x2 in uprights, chain on both sides, 3:1 on two Kraken X60s for 28 in of travel; stator limit raised from 40A to 80A. Arm of 2x2 box tube with large gussets, 51:1, shimmed gearbox. The post lists the arm as one Kraken X60 and also as 2 motors at 35A; the team reports it ended at 1 motor at 100A.

DRIVETRAIN
WCP Swerve X2i modules on a 29 in x 29 in frame, sized for stability when reaching high. Drive ratio 5.9:1, chosen for an average sprint of about 18 ft. Polycarbonate and plywood panels shield electronics from falling coral.

ENDGAME
Climb stows under the funnel and weighs about 3.5 lb with motor and gearbox. Surgical tubing powers a spring-loaded extension. One Kraken X60 retracts it through a 15:1 winch and a 2:1 pulley, sized to lift the robot off the ground with a wide CG tolerance.

AUTONOMOUS
Autos reuse the teleop auto-scoring logic and are built by chaining scoring commands at different reef locations. Return trips to the coral station use pre-planned trajectories from Quikplan, the team's time-optimal swerve planner. Coral count per auto unconfirmed (see code).

VISION
Two AprilTag localization modes. Reef tasks use only the closest tag plus swerve odometry, computed on the RoboRIO for low latency. Other tasks use all tags plus odometry in a particle filter on the driver station. Final alignment uses a trig method from tag distance and gyro angle, inspired by 6328; the team states the robot gets within about 1.5 cm and 1 deg of the target. The two misses came when the robot never saw the target tag, because there was no scoring lock-out for that case. Cameras mount on the elevator structure; count and type unconfirmed.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Build for automation first: the team stiffened the elevator and arm with 2x2 tube and shimmed the arm and wrist gearboxes so software could score with low backlash.
- A single-button scoring sequence picks the closest reef face from velocity lookahead and plans from the current position and velocity; the team reports 547 of 549 L2-L4 coral at its last two events.
- The open-top gripper lets coral rotate on exit, which gives a shallower L4 angle and more depth tolerance.
- A climb can be small: about 3.5 lb, with surgical tubing to extend and one Kraken X60 on a 15:1 winch and 2:1 pulley to retract.
- Gear ratios are stated with the design target: 5.9:1 drive for an 18 ft sprint, 3:1 elevator for 28 in of travel, 51:1 arm for 90-180 deg of rotation.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video · Offseason 'Quixshot' reveal and CAD (CD)

■ Study pick

3005 RoboChargers Relay

USA-TX

Local EPA 91 (#19)

56-9-2

'Relay': 3-stage chain-driven elevator carrying a 'laterator' with a coral ejector and an algae gripper

SCORING (CORAL / ALGAE) A 'laterator' carries a coral ejector and an algae gripper, per the reveal post. The coral ejector structure is mainly carbon fiber tube and MaxComposite. The OP states the laterator derailed when the robot climbed the reef. It also loosened side to side over the season, so the team shimmed it for an offseason event.	**INTAKE** Unconfirmed (see CAD). The team's posts do not describe the coral or algae intake path; replies about a funnel and coral routing come from other users.
ELEVATOR / ARM 3-stage elevator, per the reveal post. Chain drives it; the team says it chose chain mostly from familiarity. The bearing-to-gusset elevator hardstops started as Markforged Onyx parts and changed to MaxComposite to save weight. One Onyx hardstop deformed over time and was replaced.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The OP states the design leaves a 0.25 in gap between frame and bumpers to ease bumper removal. Bumper spacers, mounted at under 8 in intervals, are missing from the CAD.
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** Autonomous Award in the FIRST In Texas District Championship Apollo Division. Routines and coral count: Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Robot assembly CAD (Onshape)

LESSONS FOR STUDENTS

- Block CAD before detail CAD: the release includes the Block CAD document, and the team states its target of 90% Block CAD 1-1.5 weeks after kickoff.
- MaxComposite works much like polycarbonate if it is laser cut, or router cut with the edges melted to seal the woven layers. The team warns it delaminates under edge loading.
- 3D printed PLA sprockets can run on low-torque joints. The team sets the wall count high enough that the teeth print solid.
- Material swaps for weight: the elevator hardstops moved from Markforged Onyx to MaxComposite.
- Wear on a lateral slide: the laterator loosened side to side over the season and needed shims to stop derailing.

Onshape CAD: Robot assembly CAD (Onshape) · Robot assembly CAD (Onshape) · Block CAD (Onshape) · Reveal thread (CD) · Reveal video

Profile	180 S.P.A.M. Anemone USA-FL	Local EPA 90 (#23) 44-6

'Anemone': floor-pickup coral robot run by one driver, with automated scoring and a cage climb; code-only release

SCORING (CORAL / ALGAE) Coral scoring runs through an automated auto-score sequence. The controller rumbles when the sequence ends so the driver knows he has full control again. A pulsing rumble warns the driver when the drivetrain takes longer than normal to reach position. End effector type, motors and ratios are unconfirmed (no CAD released).	**INTAKE** Picks up coral off the floor in auto (release thread). Sensors detect when a coral is fully in, and the controller rumbles so the driver can release the intake button. Intake mechanism details are unconfirmed (no CAD released).
ELEVATOR / ARM Unconfirmed (not stated in the public release).	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release).
ENDGAME Cage climber. The controller rumbles once when the climber deploys and continuously when it latches to the cage. Deep or shallow cage is not stated; unconfirmed.	**AUTONOMOUS** Dynamic autos that track and pick up floor coral. Most moves (start to reef pole, reef pole to floor coral, floor coral to reef pole) use the DriveToPose command; PathPlanner handles only the curved move after the first coral. The team moved away from Choreo, used the previous season, because floor pickup needs flexible end points. Autos chain small commands with Triggers. Won Autonomous Awards at South Florida and in the Galileo Division.
VISION Limelight AprilTag vision and a coral detector for floor coral tracking. In simulation the team used PhotonVision's simulation for the Limelight AprilTag data and wrote a simulation layer for coral tracking. LEDs flag critical faults such as a disconnected camera or no AprilTags in view, so team members on the field can fix them before field staff close the gates.	**PUBLIC CAD / DESIGN** Code (GitHub)

LESSONS FOR STUDENTS

- Simulate before the robot exists: the team simulated all major mechanisms and vision so the driver could run practice cycles from kickoff to mid February.
- Use straight-line pose control where paths are simple: DriveToPose handles most auto moves, and PathPlanner is kept for the one curved move after the first coral.
- Build autos from Triggers that move the robot between states (intaking, scoring), so teleop automations can be reused without asProxy() calls.
- Give the driver feedback through rumble: coral fully held, climber deployed and latched, auto-score done, and a pulse when alignment takes too long.
- Use LEDs to show critical faults (camera not connected, mechanism out of position, no AprilTags) to team members on the field before field staff close the gates.

Code (GitHub): Code (GitHub) · 2025 code release (CD) · Code (GitHub) · DriveToPose command (GitHub) · Behind the Bumpers (video) · Robot reveal (video)

■ Study pick

5990 TRIGON TRIDOG

Israel

Local EPA 90 (#22)

46-13-2

'TRIDOG': ground coral intake (about 7 kg); elevator on custom-machined bearing blocks; 4 L4 coral auto

SCORING (CORAL / ALGAE) Scores coral on L4, as the 4 L4 autos show. End effector type and algae handling unconfirmed (see CAD and binder).	**INTAKE** Ground coral intake, about 7 kg (about 15.5 lb) after last-minute weight cuts. The weight includes its motors: 2 Falcons and one Kraken X60.
ELEVATOR / ARM Elevator on custom bearing blocks, based on 1678's 2023 bearing blocks and used by the team for two seasons. The team knew in the first week of the season it would use them and started machining them early. Each block needs two setups; a friend of the team machined the second setup for tighter tolerance.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD and binder).	**AUTONOMOUS** 4 coral ground intake auto, run as a 4 L4 auto at the ISR District Championship. The team also reports a 10 coral auto as an alliance with 1690.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Machined bearing blocks can be planned early: the team knew in week one it would reuse its 1678-based design and started machining early.
- Split a precision part across shops: the team machined the first setup and a friend of the team did the second for tighter tolerance.
- Weigh each subsystem: the ground intake is about 7 kg with 2 Falcons and a Kraken X60, and the team recalls the full robot about 0.5 lb under the limit at its last inspection.
- A ground intake auto scored 4 coral on L4 at the District Championship.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Tech binder (Google Slides) · Code (GitHub) · Reveal thread (CD) · Behind the Bumpers video (FUN Robotics Network)

■ Study pick

7407 Wired Boars Griptide

USA-CT

Local EPA 90 (#24)

55-13

'Griptide': coral robot, coral station intake, 3-stage elevator, rotating end effector for L1-L4; swerve; algae attempts dropped

SCORING (CORAL / ALGAE) Rotating end effector for L4, L3, L2 and L1 coral and for intaking (January plan, team post 2025-01-22). After Waterbury (week 2) the team planned a new end effector with a larger gripper and 2 separate Kraken X44s, for algae and supercycles. The team later states the end effector ended up different from that plan. The team states algae did not work at Hartford: the wrist swing and elevator jerk were too much even with a larger reduction on the algae motor. A second algae version added at DCMP pinched algae better but could not pull it off the reef, so the robot stayed coral-only.	**INTAKE** Coral station intake; the team decided against a ground intake because of complexity and to spend effort on software. After Waterbury the team planned to move the top VersaRoller back and add a larger polycarbonate deflector, so coral would hit the polycarbonate instead of bouncing off the roller. The team was not sure the deflector was needed; the final setup is unconfirmed. For Worlds the only change was a larger upper coral shield on the intake.
ELEVATOR / ARM 3-stage elevator (team post 2025-01-22). After Waterbury the team states a 12:1 gearing gave about a 1 second extension to L4, and planned a change to 9:1 to cut that time. Coral stuck in the robot twice at DCMP; the team added a piece of 'poly' on a bungee tied to the elevator to lift coral out, and states it was not needed after that.	**DRIVETRAIN** Swerve. The team states it used L1+ gearing for acceleration (2025-01-10), and its 2026 thread states it used custom swerve code before 2026. Module model unconfirmed (see CAD).
ENDGAME January plan: deep cage climb through an 'Intake Climb' system that shares one geared rotation between the intake and the climb. The team's later posts do not state whether the robot climbed. Unconfirmed (see CAD).	**AUTONOMOUS** Autonomous Award at NE Hartford. After Waterbury the team aimed for a 3.5 or 3.75 piece auto without speed changes. The team states it did not miss a coral in auto at Worlds (Milstein Division): 42 of 42.
VISION The team worked on PhotonVision in the preseason before the 2025 game (December 2024 post). After Waterbury it planned driver feedback for auto-align, either LEDs at the goal pose or auto-score. Camera count and placement unconfirmed (see CAD).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Scope from strategy: the team put L1-L4 coral first for its point value and chose a station intake only, so that more effort went to software.
- At kickoff, groups defined 'A', 'B' and 'C' robot architectures, planned matches in 5 second increments, and played 4 mock matches before the team set its priorities.
- Test time limits features: two algae end effectors designed between events did not work, and the team states it had no time to test them.
- A coral-only robot won the New England District Championship; the team states it had no major electrical, mechanical or code failure all season.

Onshape CAD: Onshape CAD · Reveal thread (CD) · 2025 build thread (CD) · Onshape CAD · Reveal video · Code (GitHub organization) · 2025 season recap video

■ Study pick

254 The Cheesy Poofs UNDERTOW

USA-CA

Local EPA 89 (#26)

42-7

'UNDERTOW': coral and algae robot; 4-bar main intake, backup coral station funnel; welded-tube wrist; no public CAD

SCORING (CORAL / ALGAE)
End effector of 1/4 in polycarbonate plates. It bolts to 1/4 in aluminum plates welded to the 2 in wrist tube; large fender washers spread bolt loads to stop the polycarbonate cracking. Breaks happened in the polycarbonate past this joint, so the team kept spare end effectors and swapped them at the bolted interface. Stud 'teeth' grip algae from below so it does not slip on the polycarbonate; the team says they interfered little, if at all, with grabbing. Motors: Unconfirmed (see the binder).

INTAKE
Two coral paths: a coral station intake fed by a polycarbonate funnel, and a main intake on a 4-bar linkage. A 1.25 in OD, 0.03 in wall aluminum tube jackshaft drives the 4-bar through custom billet hubs and a 20DP gear; the team says this stackup used many custom parts and was not serviceable. The funnel is cut from one polycarbonate sheet, with overlapping layers joined by double-sided tape. Flex wheels on the front two rollers were cut to fit the frame perimeter.

ELEVATOR / ARM
The wrist rides on a carriage of 2x1 tubes. 1/4 in plates on the carriage hold two 20 mm ID x-contact thin-section bearings in a live-axle setup. A 50T 10DP gear drives the wrist, and a CANcoder reads a magnet in the end plug at 1:1. The pivot is a 2 in OD, 1/16 in wall 6061-T6 tube with welded CNC end caps. The wrist was redesigned for East Bay and Championship; the binder shows the final version. A team correction to the binder, for a gearbox whose subsystem the post does not name: the gearbox is only about 11:1 (two spur stages), and a capstan drum of standoffs (0.5 in hex to 5.5 in effective diameter) gives the rest of the reduction. Elevator stages and motors: Unconfirmed (see the binder).

DRIVETRAIN
Drivetrain unconfirmed (see the binder). Bumpers use 1/2 in plywood. Four 2 in tall, 1/8 in thick steel strips in the rear bumper sides add ballast against tipping; the team chose wood over HDPE because it was lighter, which let the steel sit lower. CAN wires ride on top of power wires and are sleeved to each drivebase CANcoder.

ENDGAME
Climb mechanism: Unconfirmed (see the binder). The team's thread author writes that the 2025 cage sweeps through the robot, which forced subsystem placement.

AUTONOMOUS
The team used the coral station intake in autonomous at Sacramento and East Bay. At Championship it ran loading station autos only, not a backside algae auto, per the team's pit-scouting write-up. Autonomous Award at East Bay, Sacramento and the Milstein Division. Coral count: Unconfirmed (see the code and binder).

VISION
Left and right cameras pointed straight ahead with an overlapping region that sees the reef tag to about 5 in away (team reply, 2026-10-04). A team member mentions an auto-align that was slightly off in one match. Camera model: Unconfirmed (see the binder).

PUBLIC CAD / DESIGN
Technical binder (PDF)

LESSONS FOR STUDENTS

- Keep a backup game-piece path: the team says it could not predict how strategy would play out, kept the coral station intake as a failsafe for when the main intake broke, and ran it in autonomous at two regionals.
- Pick the architecture in order: the team breaks the game into cycles, chooses where to intake and score, sketches the robot as a black box, then picks subsystem styles.
- Put a 1:1 absolute encoder on a precision joint: the wrist CANcoder reads a magnet in the end plug, so no separate zeroing sensor is needed.
- Design one swap point: the end effector bolts on at a single interface, so the team replaced broken ones from spares.
- Check joints after long practice: plates bolted to the inside faces of 2x1 tubes held through Chezy Champs, then the tube ends split; the team says it would add crush tubes or end plugs in the future.

Tech binder: Technical binder (PDF) · Binder and code release (CD) · Technical binder (PDF) · Code (GitHub) · Reveal thread (CD) · Match video

■ Study pick

190 Gompei and the HERD StackUp (V1) and Redundancy (V2)

USA-MA

Local EPA 88 (#30)

68-19

Two released robots (V1 StackUp, V2 Redundancy): 4-motor elevator, star-wheel coral outtake, closing funnel, string climber

SCORING (CORAL / ALGAE) The V2 coral outtake uses star wheels. The team added them because the coral sometimes did not flip up onto L4 when slightly too low; the star wheels grab the coral lip and flip it onto L4. V2 also has an algae claw. It started with two X44 motors and ended with one X60 on the rollers, a larger claw gear ratio and a deeper 'C' shape, plus a 1.25 in grip-tape roller that is not in the CAD.	**INTAKE** Coral enters a funnel with belts that guide it. The funnel closes ('clapping funnel') to hold the coral when the robot moves in and out of the station quickly. On V2 the funnel also stores a coral passively while the robot holds algae. A photoelectric sensor (Allen-Bradley 42EF-D1MNAK-A2 RightSight) detects coral on the ramp. V2 added an over-the-bumper algae intake.
ELEVATOR / ARM Elevator. The team planned 6 motors for stow-to-L4 acceleration but ran 4 motors on both robots, because 6 would have saved about 20 ms. For V2 the team took weight out of the elevator gearbox and carriage. Motor type and stage count unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The team first lightened the front and back drive rails. For V2 it needed more weight savings, so it partial-depth pocketed the side rails after the robot was assembled. The climber-side rail was slightly bent at BattleCry at WPI.
ENDGAME String-and-grappling-hook cage climber. A 40 lb undamped pulling gas spring (Bansbach Easylift) pulls the deploy arm down; reeling the spool releases a retaining hook. A lightly sprung lock stops the arm folding back up. A redirector pulley keeps the fleet angle near constant so the string wraps the spool evenly. A spring-loaded 5-bar over-center linkage holds the hook on two billet blocks until string tension pulls it shut and releases it.	**AUTONOMOUS** Routines and piece counts unconfirmed (not stated in the public release). The team won the Autonomous Award at NE District URI and at the New England DCMP Ballard Division.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** CAD release (CD)

LESSONS FOR STUDENTS

- A passive part can fix a placement problem: star wheels grab the coral lip and flip it onto L4.
- Check what extra motors buy: the team planned a 6-motor elevator and ran 4, because 6 would have saved about 20 ms.
- One degree of freedom can do two jobs: the closing funnel holds coral and also moves out of the way for the climb.
- A string climber needs string management: a redirector sets the fleet angle, guide blocks keep string out of the gears, and an over-center latch releases the hook under tension.
- A second robot needs a weight budget: V2 found 10-15 lb for algae hardware from the funnel, elevator gearbox, carriage, 3/8 in thunder hex and pocketed drive rails.

CD thread: CAD release (CD) · CAD release (CD) · V1 'StackUp' CAD (Onshape) · V2 'Redundancy' CAD (Onshape)

■ Study pick

341 Miss Daisy Miss Daisy XXIII

USA-PA

Local EPA 88 (#28)

54-12

Miss Daisy XXIII: coral floor intake, continuous elevator, lantern-gear shoulder, 180 deg wrist, compact deep climber

SCORING (CORAL / ALGAE) Claw on a lantern-gear-driven shoulder with a 180 degree wrist. The wrist is belt driven to reduce backlash and weight on the arm; the team says it would try to remove the VersaPlanetary next time. The wrist zeros against a hard stop by current draw. A planned beam break was not fitted. Claw motors unconfirmed (see CAD).	**INTAKE** Coral floor intake. Vertical passive rollers keep the coral from resting in any orientation but sideways; a wave plate blocks one jam position. Bent ramps and the roller layout center the coral. The outer 2 passive rollers have silicone tubing and the center one does not, so the coral spins until centered and then climbs to the claw. The team states it also takes perpendicular coral. A belted reduction keeps the surface speed of the wheel sets about equal. An algae intake exists; its detail is unconfirmed (see CAD).
ELEVATOR / ARM Continuous elevator driven by rope from a spool. Aluminum arms set the rope lead angle to the spool, and a second set of redirects near the top aligns the rope with the pulleys. Early tests without the arms had the rope double up or leave the spool. Elevator pulleys are printed CF-nylon with 2 bearings each (the OP believes R4-2RS). The shoulder lantern gear came from an Onshape generator, with about 20 thousandths removed from the tooth tips. A code 'superstructure' keeps the arm from colliding with the robot and keeps the robot in acceptable states.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Compact deep climber, per the release post. The climber tensioner is printed CF-Nylon; the OP states it held up better than expected and is 'fairly certain' it was never replaced. Mechanism detail unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release). See the code repository.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Design an intake so the game piece has one stable orientation: bent ramps and passive rollers turn every coral sideways and center it before it climbs to the claw.
- Control rope lead angle on a spool: the team added aluminum redirect arms after early tests had the rope double up, and had no spooling issues in competition.
- A lantern gear can drive an arm shoulder: the team generated it in Onshape and trimmed about 20 thousandths from the tooth tips. Printed CF-nylon pulleys with 2 bearings each carry the elevator rope.
- A software 'superstructure' blocks unsafe arm and elevator states. The wrist zeros against a hard stop by current draw, so the planned beam break was left off.
- The team built three robots (Alpha, Beta, Comp), a plan inspired by 1678. The Beta robot's coral intake, algae intake and elevator needed major revisions before Comp.

Onshape CAD: Onshape CAD · CAD, code and documentation release (CD) · Onshape CAD · Code (GitHub) · Design binder (Google Drive) · Worlds strategy document (Google Drive)

■ Study pick

9450 Velocity Raptors

USA-WA

Local EPA 88 (#27)

50-17

Elevator with a differential wrist under custom control code; 3 piece L4 ground auto

SCORING (CORAL / ALGAE) Differential wrist: another user calls it a 'diffy wrist', and the OP's reply about its control does not correct that. The team wrote its own control scheme instead of using the Phoenix 6 differential control requests, which it did not know existed. The OP says the CTRE controls are likely faster and advises using them. End effector detail unconfirmed (see CAD).	**INTAKE** The February 2025 auto video is titled '3 Piece L4 Ground Auto'. Intake mechanism detail unconfirmed (see CAD).
ELEVATOR / ARM Elevator. The team says it spent off-season time on elevator optimization. Stages, motors and ratios unconfirmed (see CAD and binder).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD and binder).	**AUTONOMOUS** 3 piece L4 ground auto, shown 2025-02-11 before the Week 1 Glacier Peak event. Paths and path tool unconfirmed (see code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Check vendor libraries before writing custom control: the team built its own differential wrist control and now recommends the CTRE Phoenix 6 option.
- Match the mechanism to the problem: the team says it does not plan to use differential mechanisms as its default for every problem, and it calls them niche.
- Strategy detail: at Chezy Champs the alliance scored algae in the processor instead of the net when it was behind on raw points; 5940 suggested the idea.

Onshape CAD: Onshape CAD · Onshape CAD · Code (GitHub) · 3 piece L4 ground auto video · CAD, code and binder release (CD)

Other profiled robots

Each has public CAD, a binder, code or a detailed release thread. Sorted by Local EPA. ■ marks the study picks.

■ Study pick

1732 Hilltopper Robotics Private

USA-WI

Local EPA 86 (#34)

41-17

'Private': 4x L4 coral auto with QuestNav localization; algae handoff; CAD and code public

SCORING (CORAL / ALGAE) Scores coral on L4; the reveal title states 4 L4 coral in auto. A team member confirms an algae handoff that can also run in reverse in initial testing. Net algae was not the week 1 focus, so the handoff was not yet tuned. End effector type: Unconfirmed (see CAD).	**INTAKE** A team member says guarding was being added to keep coral out of the robot; it also helps the algae handoff. Coral intake type: Unconfirmed (see CAD).
ELEVATOR / ARM Unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** 4-coral L4 autonomous using QuestNav, per the reveal thread title. Path tool: Unconfirmed (see code).
VISION QuestNav localization, per the reveal thread title. The team posted a QuestNav write-up in a separate Chief Delphi thread. Other cameras: Unconfirmed (see code).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- QuestNav localization paired with a 4-coral L4 auto; the code is public, so the integration can be read directly.
- The team posted a QuestNav write-up on Chief Delphi by 2025-03-12; read it with the code.
- Keep game-piece paths apart: the team was adding guarding to keep coral out of the robot, and says it also helps the algae handoff.
- Three Innovation in Control Awards in 2025 (Archimedes, Wisconsin, Phantom Lakes) go with the public control code.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reveal thread (CD) · QuestNav write-up (CD)

■ Study pick

1114 Simbotics Simbot Suzuki

Canada

Local EPA 83 (#52)

43-22

Station-fed coral elevator with reef algae removal; swerve; deep cage climber added mid-season via a pivoting funnel

SCORING (CORAL / ALGAE) A manipulator on the elevator scores coral; the team describes L4 placements. An algae mechanism removes algae from the reef; the team added it before the deep climber because it was easy to add. At Niagara College the manipulator caught a reef branch as the elevator lowered after an L4 placement and lost its position.	**INTAKE** Coral from the loading station through a funnel. No floor coral or ground algae intake: the team ranked both last on its priority list. It states it used no resources on floor coral. The team states it would add ground algae pickup early next time, while the change was still small.
ELEVATOR / ARM Elevator. The team designed the energy chain into the CAD, which it does not normally do. Three aluminum brackets support the energy chain; a planned Lexan piece was never installed.	**DRIVETRAIN** Swerve; 'Drive Maneuverability (Swerve)' is first on the team's priority list. Module type: Unconfirmed (see CAD).
ENDGAME Deep cage climb, added after the Niagara College event. The funnel pivots down and the cage comes in from the side and twists into the robot. Climber pivot: 11T Kraken pinion, 11:80 then 18:80, 32.32:1 total. Plates broke at McMaster; at Provincials the chain sometimes sat on the 'wrong side' and the bumpers touched the ground.	**AUTONOMOUS** Unconfirmed (not stated in the public release); see the code repository.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write a ranked ability list and use it to decide what to skip: the team's 12-item list put floor algae and floor coral last and deferred deep climb until after the first event.
- Retrofitting a climber costs packaging: the elevator and funnel left no room for the cage, so the team made the funnel pivot down.
- Design the energy chain into the CAD: the team states this gave a working energy chain all season.
- Compare the 2025 code with the offseason command-based version in the second repository.

Onshape CAD: Onshape CAD · Onshape CAD · 2025 code (GitHub) · Offseason command-based code (GitHub)

■ Study pick

581 Blazing Bulldogs Zuma

USA-CA

Local EPA 82 (#59)

33-16

'Zuma' (Champs): 2-stage elevator, slip-ring arm with unlimited rotation, claw, ground coral intake with handoff; swerve

SCORING (CORAL / ALGAE) Claw driven by a Kraken X60 on a 12:1 reduction, with 3 in 30A durometer wheels on dead axles. Printed backstops add algae contact area and act as shims for coral-to-roller interference. A Banner sensor with retro-reflective tape detects coral and algae; the code auto-scores L2-L4 coral and algae in the net and processor. Changed for Champs: the team pushed the sides of the throat further out to hold algae without changing the coral grip. The team states it had made 4 claws by its Regional events and 6 claws by the Championship.	**INTAKE** Ground coral intake. The design lead states the team chose a 1323-style intake after seeing it at SVR, to cut mechanical development time and keep time for software and drive practice. The team had heard that wheel placement on 1690/2056-style intakes caused dead zones. Pivot: Kraken X44 on a 69.5:1 reduction, on a steel 10DP gear. Rollers: Kraken X44 on 5:1, in WCP SRPP plates with a Banner sensor. Changed before Champs: the top roller went from a tube roller to a hex shaft with flex wheels; the bottom roller is Cat-tongue wrapped aluminum.
ELEVATOR / ARM 2-stage sideways cascading elevator on 2 Kraken X60 at 5:1, with a 1 x 1 in 1/16 in wall diagonal brace. The carriage arm runs on one Kraken X44 at 9:1 through a custom 90T 10DP gear and a 6 in Kaydon X-contact bearing. A 6-channel slip ring allows unlimited rotation: 4 channels power the claw and 2 carry CAN.	**DRIVETRAIN** 29 x 29 in frame perimeter of 1/8 in wall aluminum tube. Four WCP X2i swerve modules with X3 gearing and eight Kraken X60. Two custom swerve covers carry Limelight mounts facing the reef. A dynamic cover of surgical tubing and rope stops coral lodging under the elevator.
ENDGAME Cage climber deployed by a torsion spring, with latches that lock the cage in. The cage wheels went from 3 in to 4 in 30A durometer to speed up cage acquisition. A Maxplanet at 9:1 winches down, and a WCP ratchet plate prevents unspooling. A CANrange behind the latch starts the climb when it detects the cage. To deploy, the winch runs until the rope fully unspools and wraps the other way; a CANcoder reads the deploy angle, and the winch then runs to a maximum angle after the cage is grabbed. Deep or shallow cage is not stated in the release.	**AUTONOMOUS** Autos run on Trailblazer, the team's own tool: segments of points, actions and constraints, followed by an adaptive pure-pursuit controller. Reusable AutoBlocks hold the scoring and intaking steps. At Champs the robot ran only a 4-piece auto: the preload plus the three prestaged lollipops, found with vision.
VISION 4 Limelights, run by one state machine that switches all cameras between game-piece detection and tag filtering. A Limelight 4 runs a CPU ML model to find prestaged algae and so locate the prestaged coral. A Limelight 3 color pipeline finds coral at handoff. The team states that camera faults degrade automation without stopping scoring.	**PUBLIC CAD / DESIGN** Champs robot CAD (Onshape)

LESSONS FOR STUDENTS

- Rebuild or modify: the team listed the Regional robot's faults (high CG when scoring, algae slipping, narrow intake, slow climb) and chose a 9-day rebuild over added subsystems.
- An arm with unlimited rotation needs motion planning. The code runs A* over a graph of safe superstructure states, with edge costs from arm angle and elevator height changes.
- Vision can replace a centering mechanism: a Limelight 3 reads coral Tx and an interpolation table sets the arm angle at handoff.
- Copy a proven mechanism to save time: the team took a 1323-style ground intake so that the rebuild time went to software and drive practice.
- Both robots are public: the Regional robot, whose first iteration was designed, built and wired in 17 days, and the Champs rebuild can be compared for the same game.

Onshape CAD: Champs robot CAD (Onshape) · 2025 CAD and code release (CD) · Champs robot CAD (Onshape) · Champs robot code (GitHub) · Regional robot CAD (Onshape) · Regional robot code (GitHub) · Behind the Bumpers video · Champs robot reveal (CD) · Champs robot reveal video

■ Study pick

6800 Valor Deadlock

USA-TX

Local EPA 82 (#57)

59-10

'Deadlock': 3-stage elevator; funnel; 3 non-drive motors pre-Worlds, 5 at Worlds; swerve

SCORING (CORAL / ALGAE) The pre-Worlds scoring mechanism used 1 motor; the Worlds version uses 2. The Worlds mechanism is cut from 1/8 in SRPP, with added flanges to stiffen it. The team states the extra motor added little weight.	**INTAKE** The pre-Worlds version focused on simple, effective human loading. The Worlds version focused on improving ground algae and coral options and added a motor to power the funnel. Ground intake mechanism detail: Unconfirmed (see CAD).
ELEVATOR / ARM 3-stage elevator with 1 motor, on the robot from the start to reduce degrees of freedom for controls. A tie rod runs from half height on the elevator stationary tubes to the 2 x 2 chassis frame rail. A planned X pair of tie rods was not built.	**DRIVETRAIN** Swerve; the team counts its mechanism motors as 'other than swerve motors'. Module type and ratio: Unconfirmed (see CAD).
ENDGAME 1 climber motor in both variants. Cage type (deep or shallow): Unconfirmed (see CAD).	**AUTONOMOUS** 3-piece auto with L4 coral. At the state championship the robot scored 50 of 51 L4 coral in auto; the team gives a 98% success rate. Autonomous Award at Manor.
VISION Unconfirmed (not stated in the public release; see the Tech Packet).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Score everything with few actuators: 3 motors besides swerve before Worlds (elevator, scoring mechanism, climber).
- Pick an architecture that reduces controls work: the 3-stage elevator was chosen to reduce degrees of freedom.
- Compare two versions of one robot: the Worlds variant added a second scoring motor and a funnel motor for ground algae and coral options.
- Material choice: the Worlds scoring mechanism is 1/8 in SRPP with added flanges for stiffness.
- Plans change during the build: a planned X pair of tie rods became one tie rod from the elevator tubes to the frame rail.

Onshape CAD: Onshape CAD · CAD release thread (CD) · Onshape CAD · Tech Packet (Google Drive) · Behind the Bumpers video · Reveal thread (CD)

■ Study pick

9496 LYNK Chimney Rock (Rocky)

USA-NC

Local EPA 81 (#69)

72-7-2

'Chimney Rock' (Rocky): low-rider chassis on SDS L2 swerve; passive gas-strut algae manipulator; reef AprilTag coral alignment

SCORING (CORAL / ALGAE) Final coral and algae end effector unconfirmed (see CAD). In week 1 the team prototyped a 'mortar style' coral shooter and rejected it. Shots were inconsistent; the physical angle did not change backspin; motor speed changed backspin but added inconsistency. The team also judged a hard coral shot unsafe for referees and human players.	**INTAKE** The algae manipulator deploys passively on a gas strut. The team linked a 25 lb, 4 in strut and states that strut force only sets deploy speed. After deployment, two polycarbonate pieces press on the aluminum extrusion to hold the manipulator in place. Coral intake method unconfirmed (see CAD); the January goal list set coral station pickup in under 1/2 second and listed ground coral intake as a non-goal.
ELEVATOR / ARM Elevator or arm unconfirmed (see CAD). A 2026 team post describes the 2025 robot as '2[1]dof' in a statement that the team keeps degrees of freedom low. A team reply in July 2025 mentions a handoff channel whose gap matches the CAD.	**DRIVETRAIN** Swerve with SDS modules at the L2 ratio, per the team's bumper post; module model unconfirmed (see CAD). The team calls the chassis a low rider and states it keeps algae from getting under the robot. The team runs its own Kraken Swerve Base code for CTRE hardware, with CANivore, FOC and an optional 250 Hz odometry thread.
ENDGAME Unconfirmed (not stated in the public release). The January goal list named a deep cage climb and listed the shallow cage as a non-goal.	**AUTONOMOUS** Unconfirmed (not stated in the public release). The January goal list set 3 coral in auto. The 2025 code post links PathPlanner and Choreo; the tool used for the final autos is unconfirmed.
VISION Vision is used mainly to align coral scoring to the reef AprilTags, per a team mentor. The mentor states that only the reef branch position relative to each face tag matters, so the team did not calibrate a field map with WPICal. PhotonVision is linked in the 2025 code post.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test a risky idea in week 1 and record the result: the 'mortar style' coral shooter was inconsistent, and the team also dropped it for referee and human player safety.
- Fast odometry has a cost: the Kraken Swerve Base adds an optional 250 Hz odometry thread, and the team says to use it only with the swerve bus on a CANivore because it raises CAN utilization.
- Chassis height is a game-piece decision: the team states its low-rider chassis keeps algae from getting under the robot.
- A passive mechanism can save a motor: the algae manipulator deploys on a gas strut and is held by polycarbonate pieces pressing on the extrusion.
- Bumper build method: layered floor-tile foam glued with contact cement replaced pool noodles, and the team irons numbers directly onto fabric stapled over the new foam.

Onshape CAD: Onshape CAD · Onshape CAD · 2025 Build Log (CD) · 2025 robot code (GitHub)

■ Study pick

971 Spartan Robotics Fiddler

USA-CA

Local EPA 80 (#84)

30-18

'Fiddler': swerve after tank in 2023-2024; stiffened elevator with arm and wrist; funnel end effector chosen over suction

SCORING (CORAL / ALGAE) Funnel end effector. The team evaluated suction but picked the funnel because it did not need a wrist at first, was reliable for algae, and had partial ground intake capability. The team later decided a wrist was needed and did not revisit suction.	**INTAKE** The team says it was committed to a human player intake early in the season. The funnel end effector had partial ground intake capability. Final intake setup unconfirmed (see CAD).
ELEVATOR / ARM Elevator carrying an arm assembly with a wrist; the team says the arm assembly weighed a lot. The elevator started as a stock Greyt elevator with two short stiffening rods and bottom gusset plates. The rods were lengthened after driving, partly inspired by Orbit. A lighter Championship rebuild was too flimsy, so the team thickened the top of the base, reduced lightening on the verticals and added U-shaped constraining plates.	**DRIVETRAIN** Swerve. A team member says defense in 2023 and 2024 drove the change from tank drive. The software team wrote custom swerve code from the ground up. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed in the thread. A user outside the team quotes the line 'Projected a ray onto the floor to calculate the distance of the coral' without naming its source; no team answer appears. See the technical documentation.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A lighter elevator can cost stiffness: the Championship rebuild focused on low weight, was too flimsy to use, and the team added material back.
- U-shaped constraining plates and longer stiffening rods are among the fixes the team lists for elevator flex under a heavy arm.
- End effector selection: the team chose a funnel over suction partly to avoid a wrist, then added a wrist later.
- Moving from tank to swerve took custom swerve code written from scratch, per the team.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Technical documentation (PDF) · Reveal thread (CD) · Reveal video

Profile

2630 Thunderbolts Atlas

Israel

Local EPA 80 (#86)

47-16-1

'Atlas': L4 coral scorer with algae into the net; 4 L4 coral auto in 14.3 s; layout in CAD and binder

SCORING (CORAL / ALGAE) Scores coral on L4, as shown in the 4 L4 auto. Algae scoring into the net was added by 2025-03-01, per the OP. End effector detail unconfirmed (see CAD and binder).	**INTAKE** Unconfirmed (see CAD and binder).
ELEVATOR / ARM Superstructure not described by the team in the thread (see CAD and binder). The team confirms a main shaft is a fixed axle held at its ends with snap clips.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD and binder).	**AUTONOMOUS** 4 L4 coral auto: 14.3 seconds from auto start to release of the last coral. It runs on the far side so it is compatible with other teams' 4-5 coral autos. The OP said in March that the first coral sometimes missed.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Plan auto to fit partners: the team ran its 4 L4 auto on the far side so other teams' 4-5 coral autos could run too.
- A fixed main axle retained by snap clips at the ends is the team's choice for the main shaft; find it in the CAD.
- The tech binder sits next to the CAD, so the design reasons can be read with the model.

Onshape CAD: Onshape CAD · CAD and tech binder release (CD) · Onshape CAD · Tech binder (Google Drive) · Reveal thread (CD) · 4 L4 coral auto video

■ Study pick

3467 Windham Windup

USA-NH

Local EPA 79 (#89)

49-19-1

Elevator with coral claw; algae in horns or a lower claw; deep cage climber; Mk4i swerve; no floor intake

SCORING (CORAL / ALGAE) Coral claw with rollers. During the UNH event the team changed the claw to catch the coral as it came through. From QM 61 through playoffs it had zero elevator jams from coral sticking out the back of the claw. Algae is held in 'horns' or a lower claw. After the first district event the team added algae scoring into the net and a better coral hard stop. Before DCMP the net score became one motion: the robot rises and spits, so it does not hook on the net.	**INTAKE** No floor intake for coral or algae; the team states this hurt its play at the highest level. Coral enters the claw from a ramp. The ramp sagged and let coral catch and turn vertical, so the team moved it about an inch closer to the claw. Before DCMP the team made the horns the only way to take algae off the reef, after it practiced processor scoring from the horns.
ELEVATOR / ARM Elevator. Each mode had its own height setpoints, and at UNH the single driver switched between three modes: coral, algae in horns and algae in lower claw. Before DCMP the code used the robot's global pose to send the elevator to the algae height for the reef face in front of it. Holding the algae button moves the elevator straight from coral scoring height to algae pickup height. Elevator motors and rigging: Unconfirmed (see CAD).	**DRIVETRAIN** Swerve on Mk4i modules: in November 2025 the team states it ran Mk4i for the last three years. The wheels were MK4 billet wheels with black neoprene tread, replaced by molded TPU tread after the season.
ENDGAME Cage climber with a climber gearbox and a cage grabber. Climber failures cost ranking points and matches at UNH, and the team planned design changes to both parts before DCMP. The team's DCMP post refers to its alliance hanging from the deep cages.	**AUTONOMOUS** 3-coral auto. During the UNH event the team added code that watches the claw rollers for current spikes. On a spike the robot ends the command and restarts it, which jostles the coral so auto can continue. The team states the auto ran worse at the World Championship than at DCMP, largely because of automation changes made without enough time or a detailed plan.
VISION Camera hardware: Unconfirmed (not stated in the public release). The code uses the robot's global pose to pick the elevator height for each reef face. The team runs auto-align code and planned before DCMP to make it handle coral between the robot and the reef.	**PUBLIC CAD / DESIGN** 2025 robot CAD (Onshape)

LESSONS FOR STUDENTS

- Detect jams from motor current: a current spike in the claw rollers ends and restarts the auto command, which jostles the coral free.
- Close geometry gaps where a game piece changes path: a sagging ramp let coral turn and jam until the team moved it about an inch closer to the claw.
- Reduce driver modes with automation: always using the horns for reef algae and pose-based elevator heights replaced three manual modes before DCMP.
- Plan changes between events: the team states its 3-coral auto regressed at Championship after automation changes made without a detailed plan.
- Make CAN faults easy to isolate: inline toggleable terminators (Thrifty CANjunction) cut diagnosis time in 2025.

Onshape CAD: 2025 robot CAD (Onshape) · 2025 robot CAD (Onshape) · 2025 build blog (CD) · 2026 build blog with 2025 review (CD)

■ Study pick

2930 Sonic Squirrels Holocentrus

USA-WA

Local EPA 78 (#100)

46-22

'Holocentrus': elevator robot; 100+ short Choreo paths; PhotonVision auto-align on 4 global-shutter cameras

SCORING (CORAL / ALGAE) PhotonVision aligns the robot to AprilTags; PathPlanner works with the Choreo paths and localization data. The team calls the result an almost fully automated scoring routine from anywhere on the field. End effector type, motors and ratios: Unconfirmed (see CAD).	**INTAKE** Two of the four main cameras are for coral station detection. Intake mechanism and ground intake: Unconfirmed (see CAD).
ELEVATOR / ARM Elevator superstructure with many self-collision points, especially when moving under the elevator. The code divides the robot into sectors and moves through each one safely. Subsystem logic checks the robot state periodically and acts on it. Stages and motors: Unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME A ThriftyCam helps the driver align to the cage. Climb type (deep or shallow) and mechanism: Unconfirmed (see CAD).	**AUTONOMOUS** Over 100 short Choreo paths let the robot run autos that the drive team writes. PathPlanner works with the Choreo paths and vision localization. Autonomous Award at Glacier Peak. Coral count per auto: Unconfirmed (see code).
VISION PhotonVision on two Orange Pi 5 coprocessors, with four AR0234 global-shutter USB 3.0 cameras: two for the reef, two for the coral station. A fifth camera, a ThriftyCam, helps the driver align to the cage. The team tested positions on an old robot against a mock reef and chose crossed views to lower AprilTag ambiguity. Two cameras did not run on USB 3.0; the team member does not recall bandwidth problems.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Build autos from a library of short paths: over 100 Choreo paths let the drive team write routines.
- Handle self-collision in software: the code splits the robot into sectors and moves the mechanism through each safely.
- Choose camera positions by test: the team mounted cameras on an old robot facing a mock reef, and chose crossed views to lower AprilTag ambiguity.
- Assign cameras to tasks: two for the reef, two for the coral station, and one ThriftyCam for cage alignment.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub)

■ Study pick

695 Bison Robotics Goldfish

USA-OH

Local EPA 76 (#124)

26-14

'Goldfish': elevator coral scorer, ramp-fed pivoting arm, modified Everybot climber; swerve; reef-repulsion auto-align

SCORING (CORAL / ALGAE) The coral mechanism, which the team calls the 'coralizer', is mostly 3D printed; the team states this made spares and replacements easier. The release states scoring is fully autonomous. Motors and ratios: Unconfirmed (see CAD).	**INTAKE** Feeder station intake through a ramp and an indexer. The ramp and indexer pivot with a combined arm that also dislodges algae. Feeder station pickup can run autonomously. No ground intake is described.
ELEVATOR / ARM Elevator built from parts of several manufacturers' elevator kits, per the team. A combined pivoting arm handles coral and algae dislodge. After the first event the team started raising the elevator earlier during auto-align. All mechanism control loops run on the motor controllers.	**DRIVETRAIN** Swerve. The OP states a full Kraken drivebase; the Kraken model is not stated. A team member states the four swerve modules ran on a CANivore bus through the team's CAN splice boards, which fit SDS or WCP modules. Module make: Unconfirmed (see CAD).
ENDGAME Modified Everybot climber; the team states it worked well for them. Deep or shallow cage: Unconfirmed (see CAD).	**AUTONOMOUS** All autos use the same auto-align system as teleop. Coral count per auto: Unconfirmed (see code).
VISION Dual Limelight 4 running MegaTag2. Odometry runs at 200 Hz with a NavX2. This localization feeds the reef auto-align.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A simple auto-align can be enough: P control on x, y and theta, plus a repulsion field that switches on only when a straight path to the target would hit the reef.
- Model a controller before coding it. The team published a Desmos proof of concept of the repulsion field that anyone can drag and animate.
- Sequence the superstructure against the drive. The code compares elevator time to setpoint with drive time, then slows or backs up to avoid hitting coral already scored.
- 3D printing most of the coral mechanism made spares and replacements easier, per the team.
- The CAN splice PCB design and Gerber files are public, with a stated parts cost of about $20 per board.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Reef repulsion model (Desmos) · Auto-align video · CAN splice PCB files (Google Drive)

■ Study pick

3128 The Aluminum Narwhals Gillbert

USA-CA

Local EPA 76 (#122)

31-13

'Gillbert': elevator robot; sensorless counter-rotating coral end effector; interchangeable L1 coral and algae intakes

SCORING (CORAL / ALGAE) Coral end effector with front and back wheel sets about one coral length apart that spin in opposite directions. At the elevator stow position the wheels run all the time; the back wheels push the coral forward and the front wheels push it back to hold it. There is no beam break or banner sensor on the end effector. A slit in the top polycarbonate plate, bent down, acts as a one-way latch; the team member does not believe it is in the CAD.	**INTAKE** Coral loads from the coral station through a funnel into the end effector. A ground coral mechanism that scores only on L1 was added during the season for champs. Because of the weight limit, the coral and algae intakes are interchangeable and are not on the robot at the same time.
ELEVATOR / ARM An elevator carries the end effector. A reef algae remover stows inside the frame and snaps out on surgical tubing to a hard-stop locked position when the elevator first rises in auto. It extends 3.25 in from the frame, the same as the bumpers. Elevator stages, motors and ratios: unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** Coral count and paths: unconfirmed (not stated in the public release). The algae remover deploys on the first elevator raise in auto. TBA lists the Autonomous Award at the San Diego Regional.
VISION Unconfirmed (not stated in the public release). A team member mentions work on auto align while the team sped up its autos.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Geometry can replace a sensor: two counter-rotating wheel sets spaced one coral length apart hold the coral, so the code needs no intake state change.
- When fast scoring threw coral into the back wheels, the team added a bent polycarbonate latch instead of a sensor.
- Weight drives architecture: the team made the coral and algae intakes interchangeable instead of carrying both.
- Review upgrades after the season: the team calls the L1 addition for champs 'a big win' and the barge algae addition 'a miss' because it cut coral scoring efficiency.
- A passive deploy: the algae remover springs out on surgical tubing the first time the elevator rises.

Onshape CAD: Onshape CAD · Onshape CAD · Technical binder (Google Doc) · Reveal thread (CD) · Reveal video

■ Study pick

5460 Strike Zone TITAN

USA-MI

Local EPA 75 (#141)

52-12-2

'TITAN': continuous elevator with a 50 mm shaft arm; rotating coral claw with a BAG motor in a carbon tube; winch climber

SCORING (CORAL / ALGAE) Rotating coral claw. A BAG motor and VersaPlanetary Lite sit inside a carbon tube. The hex output shaft drives aluminum plates and 3D prints in the claw. A custom through-bore encoder housing holds a CANcoder for closed-loop position control. Algae handling unconfirmed (see CAD).	**INTAKE** Asked about a ground intake, the OP replied 'no'. Coral intake detail unconfirmed (see CAD and binder).
ELEVATOR / ARM Continuous elevator with an arm on the carriage. The arm pivots on a 50 mm shaft on bearings bought from Amazon, the same setup as the team's 2023 arm joints, based on 971's 2018 arm. The main pivot has three bearings: two near the large pulley for the radial load and one at the shaft end to keep the joint parallel. SLS-printed Nylon 12 parts join the laser-cut pulley to the arm. The aluminum shoulder was machined from 2.5 in OD x 1.5 in ID round stock on a sponsor's CNC lathe. The arm belt has no adjustable tensioner: its path around the large pulley, the 30T pulley and two idlers was set to a slight undersize of a 650 mm (130T) belt, and the idlers add wrap on the 30T pulley to prevent skipping. The elevator belt path length does not change, so it needs no tension spring; the team tensioned it before Week 1 and did not touch it after. A cross bar on the back of the elevator stops it from racking into a parallelogram. CF springs meant to hold the first stage down were removed; the carriage and arm weigh over 20 lb.	**DRIVETRAIN** WCP X2St modules, named by the team as the mounting base for its WCP bumper mounts. Ratio and motors unconfirmed (see CAD). Bumpers weigh about 30 lb with added weight plates to lower the center of gravity; the team says this helped it stay stable at full height.
ENDGAME Winch climber. A 3D-printed latch on the winch gearbox cross shaft holds a spliced Dyneema loop around the climber foot. In climb state the shaft turns 1 rotation in reverse and the cage centering device springs out. Deep or shallow cage unconfirmed (see CAD and binder).	**AUTONOMOUS** Unconfirmed (not stated in the public release). The team calls TITAN its most complex and automated machine.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD (imported from SolidWorks)

LESSONS FOR STUDENTS

- Reuse proven joints: the 50 mm arm shaft repeats the team's 2023 design, which was based on 971's 2018 arm.
- A motor and planetary can fit inside a carbon tube to rotate a claw, with a CANcoder in a custom through-bore housing for position control.
- A belt can need no tensioner: the elevator belt path length stays constant, and the arm belt path was sized to a slight undersize of a 650 mm belt with idlers for wrap.
- Build stiffness on purpose: the team advises pre-made bearing blocks, 6 in or more between bearing centers on overlapping stages, and a force diagram. Its force diagram led to a cross bar against racking.
- The team weighed investment casting against machined parts and kept sheet metal plates and machined tube on the carriage.

Onshape CAD: Onshape CAD (imported from SolidWorks) · Reveal, CAD and binder thread (CD) · Onshape CAD (imported from SolidWorks) · Technical binder (Google Doc) · Reveal video

■ Study pick

294 Beach Cities Robotics Sacabambaspis

USA-CA

Local EPA 73 (#160)

27-18

'Sacabambaspis': station hopper; pivoting wrist with pass-through side rollers; 3-stage elevator tilted 3 deg forward

SCORING (CORAL / ALGAE)
A pivoting wrist carries the coral and algae end effector. Coral passes through on side rollers; the team chose this because the side rollers worked on its test unit. The algae manipulator also changed to side rollers so it fit in the starting configuration. The team states the robot handles coral and algae independently. Motors and ratios: Unconfirmed (see CAD).

INTAKE
Coral loads from the station into a low-powered hopper on the side opposite the scoring side. The kit-bot test phase showed the value of a wide acquisition zone under defense, so the team lowered the ramp into a hopper. No coral ground intake; the team states it could not implement the concepts it studied in its time and resources. An earlier concept had a 4-bar algae ground intake, which the team removed. Algae ground pickup on the final robot: Unconfirmed (see CAD).

ELEVATOR / ARM
3-stage elevator, chosen to reach the barge and to reduce overall robot height. The elevator moved toward the front of the robot and tilts forward 3 deg. The team states the tilt gives room to stow, reach out to the reef and leave room for the cage chain. The FUN Behind the Bumpers summary describes it as a continuous elevator.

DRIVETRAIN
Drivetrain unconfirmed (see CAD).

ENDGAME
The concept-stage layout had a climb that pulled the cage into the hopper. The team's list of final-layout changes does not remove this climb. The FUN Behind the Bumpers summary describes a powered hopper with integrated climb. Deep or shallow cage: Unconfirmed (see CAD).

AUTONOMOUS
Unconfirmed (not stated in the public release; see code).

VISION
Unconfirmed (not stated in the public release). The team states the oversized-elevator test phase showed that auto-drive routines for alignment were critical.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Set requirements before concepts. The team's four requirements (load one side and score the other, prioritize L2-L4, handle coral and algae independently, ground algae to the barge) ruled out several architectures.
- Build a test unit in parallel with the competition robot. The kit-bot phase showed that a wide acquisition zone matters under defense, and the oversized-elevator phase taught the team continuous elevators.
- Record rejected concepts and the reason for each. The post names the coral ground intake layouts the team studied on other robots and why each did not fit its requirements or resources.
- Packaging changes cascade. The end effector pushed the elevator toward the center and into the cage chain, which led to the forward move, the 3 deg tilt and the 3-stage elevator.
- Fit the drive position to the robot. The team states that independent coral and algae handling suited driving from station 1, where it cycled through playoffs.

Onshape CAD: Onshape CAD · CAD, code and binder release thread (CD) · Onshape CAD · Code (GitHub) · Technical binder (PDF) · Annual report (PDF) · Behind the Bumpers video

■ Study pick

6995 NOMAD Mariana

USA-CA

Local EPA 73 (#163)

54-16

'Mariana': elevator and pivot with a two-sided ground 'hand' for coral and algae; single-pole wheeled cage latch climb

SCORING (CORAL / ALGAE) A ground 'hand' end effector, reached after about 30 iterations of wheels and small changes. It scores coral on both sides of the robot at L3 and L4 and removes algae from the reef on both sides. The team expected 4-5 more cycles per match from it.	**INTAKE** Coral station intake at the Orange County Regional. The team switched to the ground hand for Championship because station intake suffered under heavy defense. The team thanks 9442 and 8033 for help with it.
ELEVATOR / ARM Elevator with a pivot. A Vortex motor drives a WCP rubber friction brake on an 8mm lead screw into a lathed aluminum hex hub. The hub is on the same shaft as an early pivot gear, so little force is needed. The 3D-printed pivot parts are PLA pro and carry no load; they act as spacers or axial constraints.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). Electronics mount under the robot behind a polycarbonate cover held by bolts into rivnuts; the team states access takes about 30 seconds. A team member states the team switched from REV to Kraken motors this season; the post does not say which mechanisms. Drive motors unconfirmed (see CAD).
ENDGAME Wheel-based latch climber on a single cage pole, revived from an early prototype. The earlier hook climber held for less than 3 seconds. The grabber is angled up 12 degrees so it reaches outside the bumpers and pulls the eye bolt more than an inch lower. Expected climb time dropped from 15 s to 5-7 s.	**AUTONOMOUS** Autonomous Award at the Orange County Regional; Programming Award at Beach Blitz. A separate drivebase let software test auto cycles before the main robot was ready. Routines: Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Put a holding brake early in the reduction: the friction brake acts on a shaft near the motor, so a small lead screw force holds the pivot.
- A single-pole wheeled cage grabber needs less alignment: the team states it catches a corner at almost any approach angle.
- Change geometry before adding parts: angling the existing grabber up 12 degrees fixed its reach without building a two-pole version.
- Give software a separate drivebase: the team used one to test auto cycles before the competition robot was ready.

Onshape CAD: Onshape CAD · Onshape CAD · Behind the Bumpers video · Reveal video

■ Study pick

3061 Huskie Robotics Calypso

USA-IL

Local EPA 70 (#220)

25-15

'Calypso': SDS MK4n swerve; side-to-side funnel into a four-stage elevator coral scorer; barge algae manipulator; winch climb

SCORING (CORAL / ALGAE)
Coral manipulator on the elevator scores or lightly shoots coral onto the reef. Prototype wheels were 2.25 in 40A AndyMark wheels, 1.5 in apart and 4 in inside-to-inside, chosen after intake, spit-out and hold tests. For L1 the team added a tab below the coral that hits it as it flips. Changed after Week 1: a separate algae manipulator for barge and processor, based on the Northern Lights algae removal mechanism; the team places algae in the barge and reports 100% on barge shots at Midwest.

INTAKE
A funnel feeds coral from one side of the robot to the other. At kickoff the team judged a coral ground intake not worth it; after Northern Lights it listed a possible ground intake as less important than barge algae. A horizontal-roller funnel jammed in testing, so the team began to prototype vertical rollers, inspired by 4451. At Northern Lights the funnel slowly bent and an over-tight belt caused the same fault; the planned changes for Midwest were to remove the two back wheels and to use star wheels around compliant wheels. Week 1 algae plan: a slap-down processor intake with polycarbonate tube rollers.

ELEVATOR / ARM
Four-stage, continuously rigged elevator on two Krakens, modeled on a 2023 Worlds-winning elevator. Play in the elevator made L4 inconsistent in auto at Midwest. At Championship the team found failed tube plugs and bolts in one stage, which let it parallelogram, and no bolts holding one side to its mount. The team replaced and threadlocked the hardware and upgraded failed ball joints on the elevator support tubes with hardened bolts.

DRIVETRAIN
Swerve on all SDS MK4n modules, chosen for packaging: a wide intake on one side and the elevator close to the outer rail on the other. Choreo analysis led the team to L2+ gearing for acceleration and short sprints. Code drives the robot the other way when it tips so it does not stay stuck on algae.

ENDGAME
Week 1 plan: an angled telescoping tube with a spring-deployed cage catcher; design requirements called for a shallow climb only. Changed in March after the cage stuck in the robot at Northern Lights: a 111-style climb clashed with the funnel in CAD, so the team used a design inspired by 190. Two stacked cage catchers grab one cage bar; the lower 'hand' sits on a wrist that rotates passively when the winch pulls the hook back, and a WCP ratchet plate locks it. A servo moves the funnel bottom out of the cage's path; early deploys, a snapped rope and bent limit switch mounts followed at Midwest and Championship. Deep or shallow cage unconfirmed.

AUTONOMOUS
A 3.5-coral auto by 2025-03-11, within 15 s; it ends with a sprint to the reef with the swerve modules in coast mode. After Midwest the team listed a 4-coral auto as the first priority. Path tool unconfirmed (see code).

VISION
Week 1 software plan: four cameras. Two front cameras on the elevator face inward toward each other for reef tags; two rear cameras on the swerve modules face outward. Late at Northern Lights the pose estimate drifted; the team suspected the cameras had moved and planned mounts that locate each camera exactly. Software added drive-to-pose for barge and processor. Camera and coprocessor type unconfirmed.

PUBLIC CAD / DESIGN
Onshape CAD

LESSONS FOR STUDENTS

- Design requirements set the archetype: a Week 1 event made simplicity the top priority, so the team picked a funnel and vertical elevator with 2 degrees of freedom for scoring.
- Check a borrowed mechanism in your own CAD: the team compared five existing climb designs and dropped the 111-style climb when it clashed with the funnel.
- Structural faults can look like control faults: elevator sway at Championship came from failed tube plugs, loose or missing bolts and broken ball joints.
- Camera mounts are part of the vision system: the team traced late-event pose drift to cameras that it suspected had moved.
- Log analysis closes the loop: the team's log-analyzer reported a 7.04 s average cycle over 31 cycles in two Curie matches, and a 7.8 mm average lateral reef error at coral release.

Onshape CAD: Onshape CAD · 2025 build thread (CD) · Onshape CAD · Code (GitHub) · 3061-lib (GitHub) · Testing videos (Google Drive) · log-analyzer post in preseason thread (CD) · log-analyzer tool (GitHub)

■ Study pick

6328 Mechanical Advantage

USA-MA

Local EPA 70 (#209)

44-21

Dev bot plus four competition CAD versions (25B, 25C, 25D, 25O); arm pivot; archetype otherwise unconfirmed (see CAD)

SCORING (CORAL / ALGAE) Unconfirmed (not stated in the public release; see CAD).	**INTAKE** Intake type unconfirmed (see CAD). The intake mounting plates attach to 3D-printed parts that have nylock nuts pressed into the back and are printed at high infill. The team reports no issues with them.
ELEVATOR / ARM The robot has an arm pivot. The team makes the pivot's tube plugs on a lathe and holds them with green Loctite. It reports no issues with that pivot and would add a roll pin through each plug in future, as 2056 does. Elevator: Unconfirmed (see CAD).	**DRIVETRAIN** A team reply on the 2025 build thread confirms a metal standoff runs through a printed support at the swerve bolt. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release; see CAD).	**AUTONOMOUS** Autonomous Award at the NE District UNH Event. Routines and coral count: Unconfirmed (see code).
VISION The team states its 2025 vision moved past Orange Pi coprocessors, which gave better data plus object detection and video recording. Each camera sent at least 2-3 Gb/s of raw video. The camera mounts are TPU. The team corrects rotation estimates with vision data and does not trust the gyro alone. Camera count and coprocessor model: Unconfirmed.	**PUBLIC CAD / DESIGN** 2025 build thread with CAD and code links (CD)

LESSONS FOR STUDENTS

- Compare versions. The build thread links a dev bot and four competition CAD versions (25B, 25C, 25D, 25O) from one season.
- Design printed parts to the load path. The team prints nearly every loaded part in PLA+ with 5 walls, 5 floors, 5 ceilings and 75% infill. It tries to make load-bearing parts metal, or to add a metal support, such as a standoff inside the printed swerve bolt support.
- Plan for wear. The team describes printed pulleys as wear items to replace about once an event.
- Code performance is a design choice. The team states its 2025 code used the serial garbage collector with settings taken from 254. The code needs about 45 s to warm up, and a dashboard alert tells the drive team when it is safe to enable.
- Keep a design-review record. A team mentor posted an album of the design-feedback screenshots sent to the CAD students during the season.

CD thread: 2025 build thread with CAD and code links (CD) · 2025 build thread with CAD and code links (CD) · 25A Dev Bot CAD (Onshape) · 25B Comp CAD (Onshape) · 25C Comp CAD (Onshape) · 25D Comp CAD (Onshape) · 25O Comp CAD (Onshape) · Code (GitHub) · 2025 offseason code (GitHub) · 2025 design feedback album (Google Photos) · SystemCore Alpha thread, 2025 vision and code notes (CD) · 2025 build thread, conference recordings post (CD)

Profile	3414 Hackbots Levi USA-MI	Local EPA 69 (#233) 34-26

'Levi': elevator with a fixed-angle coral end effector and a separate algae mechanism; swerve

SCORING (CORAL / ALGAE) Stationary coral end effector set at 35 deg from the floor. At the first competition it sat at about 50 deg, chosen to balance L2/L3 and L4 scoring. The robot has separate coral and algae mechanisms. Algae mechanism detail: Unconfirmed (see CAD).	**INTAKE** Foam coral blockers are on the robot, per the OP. Coral and algae intake method: Unconfirmed (see CAD).
ELEVATOR / ARM Elevator that reaches barge height, per a team member's reply. A cable chain carries wiring up to the coral and algae mechanisms. Team members state the elevator crushed the chain when coral fell into the front of the robot, and the chain swings when the elevator moves. Elevator motors and rigging: Unconfirmed (see CAD).	**DRIVETRAIN** Swerve, per the OP's description of the cable chain folding against the swerve module. Module type and ratio: Unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** Routines and coral count: Unconfirmed (not stated in the public release). The release post links Autopilot, a holonomic motion control project the team developed this season.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** CAD release thread (CD)

LESSONS FOR STUDENTS

- Test cable chain routing with game pieces present: the first routing worked in testing but broke at competition when coral and the coral blockers hit it.
- The revised routing used existing geometry, the swerve module and foam coral blockers, to guide the fold, with a switched bend at the end and an outer polycarbonate guard. The OP calls the result mostly luck.
- A fixed-angle coral end effector avoids a pivot. The team moved the angle from about 50 deg to 35 deg after its first competition.
- Electrical standards as a document: the team's 2024-25 Electrical Manual covers connectors, CAN and soldering practice.

CD thread: CAD release thread (CD) · CAD release thread (CD) · Onshape CAD · Autopilot motion control thread (CD)

Profile	5419 Berkelium Argonaut USA-CA	Local EPA 69 (#227) 27-17

'Argonaut': coral-focused robot with elevator, pivot and climb; auto-align to each reef branch

SCORING (CORAL / ALGAE) Coral end effector. The release post has an 'End Effector' heading with no text description. The team states it focused on getting coral onto the reef rather than doing everything. Algae handling unconfirmed (see CAD).	**INTAKE** Unconfirmed (not stated in the public release). Station or ground coral intake: see CAD and binder.
ELEVATOR / ARM Elevator with a pivot; the release names both as subsystems. Stages, motors and ratios unconfirmed (see CAD and binder).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME The release names a climb subsystem. Deep or shallow cage unconfirmed (see CAD and binder).	**AUTONOMOUS** The release lists an autonomous routine that scores 3 L4 coral. Paths and path tool unconfirmed (see code).
VISION Camera hardware unconfirmed (not stated in the release text). The release lists auto-align to each branch of the reef.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Scope is a design decision: the team states it chose a simpler, coral-focused robot to improve reliability and consistency.
- The release lists a 3 L4 coral auto and auto-align to each reef branch; the public code is the place to study both.
- CAD, code and a technical binder are linked from one post, so a student can cross-check one mechanism across all three.

Onshape CAD: Onshape CAD · CAD and code release thread (CD) · Onshape CAD · Code (GitHub) · Technical binder (Google Drive) · Season recap video

■ Study pick

1156 Under Control Ultralisk

Brazil

Local EPA 68 (#243)

32-12

'Ultralisk': swerve with an always-on heading PID set by the right-stick angle; mechanisms unconfirmed (see CAD and binder)

SCORING (CORAL / ALGAE) Unconfirmed (see CAD and binder).	**INTAKE** The team set a design target for the intake: move coral from the Coral Station to the end effector in under 2 s while the robot moves. Ground intake unconfirmed (see CAD).
ELEVATOR / ARM Unconfirmed (see CAD and binder).	**DRIVETRAIN** Swerve, per a team reply in the release thread. The drive always runs an orientation PID. The driver points the right stick at the wanted heading, and atan2 gives the target angle. A bumper button lets the driver spin the robot freely against defense. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD and binder).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION A team reply mentions object detection and driver buttons that cover its false positives and false negatives. Camera and pipeline unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Heading as a position command: the right stick sets a target heading through atan2, and an always-on orientation PID holds it. The team states this prevents heading drift.
- Keep a manual escape. A bumper button gives free spin for defense; the team says spinning by circling the stick 'felt weird'.
- Plan the build by milestone: archetype decided at the end of Week 1, a master sketch that weekend, and a target of a mechanically complete robot by the first week of February. The team calls these the milestones it usually tries to reach.
- Give each subsystem a measurable target, for example coral from the station to the end effector in under 2 s while moving.

Onshape CAD: Onshape CAD · CAD, code and tech binder release (CD) · Onshape CAD · Code (GitHub) · Technical binder (Google Drive) · Heading control code (GitHub) · 2025 build season schedule (Google Sheets)

■ Study pick

8159 Golden Horn Serpentheim

Turkiye

Local EPA 66 (#295)

32-11

'Serpentheim': coral funnel and a separate algae system; two cameras; deep cage climb added after Week 2

SCORING (CORAL / ALGAE) Separate coral and algae scoring systems. The team says it designed the algae system so it did not interfere with coral scoring. Mechanism detail unconfirmed (see CAD and binder).	**INTAKE** Coral funnel and an algae intake. The team iterated both after Week 2 to cut cycle times. Ground coral intake unconfirmed (see CAD).
ELEVATOR / ARM Unconfirmed (see CAD and binder).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Deep cage climb, added after the Week 2 event.	**AUTONOMOUS** The team says frequent tag localization improves alignment and autos. Coral count unconfirmed (see code).
VISION Two cameras; the forum question the team answered calls them Limelights. A team mentor gives three reasons: see AprilTags when fully aligned to a branch, see at least one tag at all times, and keep scoring if one camera fails. The second camera mount is missing from the CAD assembly.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Design from a game prediction: the team expected full reefs at Championship and made net algae a separate, non-interfering system.
- Two cameras widen the view: the team wants tags in view when fully aligned to a branch, and at least one tag in view all match.
- Camera redundancy lets the robot keep scoring if one mount breaks.
- The strategy document is public, so the design choices can be traced back to the stated goals.

Onshape CAD: Onshape CAD · CAD, code and strategy release (CD) · Onshape CAD · Code (GitHub) · Game analysis and strategy (Google Doc) · Technical binder (Google Drive) · Robot showcase video

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■ Study pick

4795 Eastbots

USA-NC

Local EPA 64 (#314)

41-19

Elevator and chain-driven arm with a motorized wrist (added for DCMP); passive coral funnel; WCP Swerve X Flipped swerve

SCORING (CORAL / ALGAE) Roller end effector with 2 in rollers on a wrist at the end of the arm; motors unconfirmed (see CAD). Changed before DCMP: a motorized wrist replaced the VFB so the team can set the outtake angle and score algae in the net. The team named a mismatch between the funnel feed angle and the rollers as a cycle-time bottleneck. A V3 end effector with three contact points gripped coral better but struggled to intake algae in manual testing, so the team planned to keep V2 with 1/8 in polycarbonate plates in place of 1/4 in.	**INTAKE** Passive funnel that accepts coral from all directions at the coral station. After the district events the team added a top plate to stop fast-fed coral passing through a gap in the arm. A MAXTube on the bottom plate stops coral bouncing out. The March 2025 post states the robot no longer gets an over-the-bumper algae intake; ground coral intake unconfirmed (see CAD).
ELEVATOR / ARM Elevator with a chain-driven arm. The end effector passes through the elevator; the team calls it a pass-through mechanism. Custom arm and wrist gearboxes reduce backlash and weight high on the robot; one custom gearbox holds both the wrist motor and the intake motor. Changed before DCMP: the VFB chain became a belt, the arm encoder moved to the top MAXSpline so arm position survives a skipped chain, and steel cables with a turnbuckle tension the elevator. Code holds arm and elevator setpoints for L1-L4.	**DRIVETRAIN** Swerve with WCP Swerve X Flipped modules. Changed in March 2025 from tube mount to corner mount for a wider wheelbase and less frame rail weight. Frame rails went from 3/16 in to 1/8 in wall. Drive and steer motors unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Two AprilTag cameras compute the robot's position. Code includes automated commands that align to and score on the reef and barge. Camera models unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One gearbox for two motors: the team states that a custom gearbox holding both the wrist and intake motors cut weight at the top of the elevator.
- Read the arm angle after the chain: moving the arm encoder to the top MAXSpline keeps the arm position known if the arm chain skips.
- Brace around a pass-through: crossed cables are not possible with the pass-through mechanism, so the team ran steel cables with a turnbuckle on the side opposite the A-frame.
- Cut weight from wall thickness once loads are known: frame rails went from 3/16 in to 1/8 in and elevator uprights from 1/8 in to 1/16 in.
- Test a redesign before it goes on the robot: the V3 end effector gripped coral better but struggled with algae, so the team planned to keep V2 with thinner plates.

Onshape CAD: Onshape CAD · Onshape CAD · 2025 build thread (CD) · 2025 CAD and code release post (CD) · Team GitHub

Profile	418 Purple Haze USA-TX	Local EPA 62 (#362) 36-24

Sideways 1-stage elevator + arm ('Armavator'); 'Thagomizer' coral scorer/algae remover; side-biased funnel; Everybot-style climber

SCORING (CORAL / ALGAE) 'Thagomizer' outtake. The team packaged coral scoring and algae removal on one shaft to keep it light. Motors and ratios unconfirmed (see CAD).	**INTAKE** Coral station funnel, biased to one side to make room for the climber and to line up with the outtake. The first version jammed; the team removed the side wheels and bottom roller and changed the geometry. Ground intake prototypes did not go well in the first week, and the posts through build week 5 describe no ground intake.
ELEVATOR / ARM 'Armavator': a sideways one-stage elevator with an arm. The sideways layout biases the center of gravity for the climb. The team moved from a two-stage elevator to elevator plus arm to package algae removal with coral scoring. The moving assembly carries 4 beam break sensors, 1 NEO with a SPARK MAX and 1 Kraken X60.	**DRIVETRAIN** Swerve X2i modules with Kraken X60 drive and steering motors; the team's first season on CTRE hardware. 29 in x 29 in frame for a larger coral station intake area, with a 1/8 in aluminum belly pan. About 52.5 lb without bumpers and about 60 lb with bumpers. Bumpers use stacked EVA foam floor tiles with a 1678-style mount.
ENDGAME Cage climber based on the Everybot climber, with added guides that funnel the cage in. Tensioned elastic gives the hook one-way movement to catch the cage. Deep climb was in the week 2 archetype; the final climb type is unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release). The build thread mentions auto-alignment work but names no cameras.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A sideways elevator can shift the center of gravity to one side to help the climb.
- An elevator plus arm can put two jobs on one mechanism, at a wiring cost: 4 beam breaks and 2 motors ride on the moving assembly.
- Check pulley tooth counts in CAD against the parts: the team ordered wrong belts because modeled pulleys had extra teeth.
- More rollers do not always help a funnel: the team removed the side wheels and bottom roller and edited the geometry to stop the jams.

Onshape CAD: Onshape CAD · Onshape CAD · 2025 build thread (CD)

Profile	1153 Timberwolves Lagotto USA-MA	Local EPA 62 (#369) 46-24

'Lagotto': 1-stage elevator scoring coral L1-L3 and algae in processor and barge; ground algae intake; deep climb

SCORING (CORAL / ALGAE) Scores coral on L1, L2 and L3, and algae into the processor and the barge, per the release post. End effector detail unconfirmed (see CAD).	**INTAKE** Ground algae intake. Coral intake type unconfirmed (see CAD).
ELEVATOR / ARM 1-stage elevator. Motors and ratios unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Deep cage climb; the team reports it takes under 15 seconds. Mechanism unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release). The team won its first Autonomous Award at the New England DCMP; see the code.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A 1-stage elevator robot can cover L1-L3 coral plus both algae goals; compare its reach with multi-stage designs in the CAD.
- The release pairs a ground algae intake with a deep climb that the team times at under 15 seconds.
- The code link points to a branch named 'DCMP'. The post does not say which event the branch matches.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Code (GitHub) · Teaser video

■ Study pick

587 The Hedgehogs

USA-NC

Local EPA 61 (#372)

38-20-2

Hopper-fed coral scorer on an elevator with a pass-through wrist; full-width slap-down algae intake for the processor

SCORING (CORAL / ALGAE) Coral end effector on a wrist that grips the coral from the inside with compliant wheels. The team chose the inside grip because it made the design smaller and easier to handle. The team planned polycarbonate plates and planned to add a switch sensor inside to detect a held coral. Status as of the 2025-02-07 build update; check the CAD for the final version.	**INTAKE** Coral: a hopper at the coral station feeds the end effector. Tests showed a passive hopper worked as well as one with powered alignment wheels; a higher opening, an angle matched to the chute and a fully supported opening reduced bounce-outs. Algae: a slap-down intake that spans the full robot width, mounted under the hopper, picks algae off the ground for the processor.
ELEVATOR / ARM Elevator with a wrist that passes through it, so the robot loads on one side and scores on the other. The wrist rides up the elevator and reaches 8-10 inches above the carriage at full extension, for L4. It ran on chain at the time; the team was considering belt or direct gearing.	**DRIVETRAIN** 27 in x 27 in base. Printed spacers lower the drive base for a lower center of gravity. Drivetrain type unconfirmed (see CAD).
ENDGAME No climber as of 2025-02-07; the strategy did not prioritize climbing. The team was designing a shallow climb (hook on the elevator) or a deep climb with a stationary hook inspired by 4481. Final endgame unconfirmed (see CAD).	**AUTONOMOUS** Routines unconfirmed (not stated in the public release). Won the Autonomous Award at the FIRST North Carolina State Championship.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Test a hopper before adding motors: the team found a passive hopper aligned coral as well as one with powered wheels.
- Bounce-outs at the coral station dropped with a higher hopper opening, an angle that matches the chute, and a fully supported opening.
- Keep closely coupled subsystems in one CAD document: the hopper and elevator share a document so their spacing stays consistent.
- An inside-grip coral holder with compliant wheels gave a smaller end effector; moving the wheels inside the plates removed the need for wheel overlap.

Onshape CAD: Onshape CAD · Open Alliance build thread (CD) · Onshape CAD · Team GitHub

Profile	233 The Pink Team Bubbly Anemone Enemy (BAE) USA-FL	Local EPA 58 (#458) 36-17-1

'Bubbly Anemone Enemy': chain-and-Dyneema cascade elevator, coral shooter, floor algae collector, deep climber

SCORING (CORAL / ALGAE) Coral shooter, the main mechanism ('Final Coral Shooter' sub-assembly). Motors and roller layout unconfirmed (see CAD). A student CAD lead states the alpha robot was too short to score on L4.	**INTAKE** Floor algae collector, the second mechanism ('Final Algae Collector' sub-assembly). Coral intake method unconfirmed (see CAD).
ELEVATOR / ARM Cascade elevator driven by 35 chain with Dyneema rope. The mentor states every elevator part was sized for 1000 lb of force, with six #4 screws and Dyneema rated to 3000 lb. Turnbuckles with jam nuts set chain tension and were checked each match. The CAD shows 25 chain, an idea the team considered to save 1 lb; the robot ran 35 chain the entire time.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Deep climber ('Final Deep Climber' sub-assembly), on the final robot only. The mentor states it was the least reliable part of the robot because it was kept out of the alpha design.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Final robot CAD (Onshape)

LESSONS FOR STUDENTS

- Pick the lift drive from the requirements: the mentor compares belt (compact, fewer failure points), chain (repeatable under load, heavier) and rope (light, harder to rig, abrades).
- A chain cascade lift can be low-backlash: the team reports position within one encoder tick on a REV quadrature encoder.
- A cascade lift gives a predictable center of gravity and loading case, which the mentor says makes PID tuning easier.
- An alpha robot only tests what it contains: the climber was left off the alpha robot and became the least reliable mechanism.
- Keep turnbuckles tight with thread locker, jam nuts and a per-match check; the mentor also suggests safety wire.

Onshape CAD: Final robot CAD (Onshape) · 2025 CAD release (CD) · Final robot CAD (Onshape) · Alpha robot CAD (Onshape)

Profile	1710 The Ravonics Revolution Dobson USA-KS	Local EPA 57 (#483) 25-13

'Dobson': elevator robot for coral L1-L4; algae claw for reef algae and barge; ground algae intake added for Champs

SCORING (CORAL / ALGAE) Scores coral on L1-L4 and algae in the barge; the team also practiced processor scoring. The team's match logs list coral per level. In Qual 15 it went 8 for 9 on L4 in teleop; the Qual 50 log lists 9 on L4. An algae claw takes algae off the reef; in one match the robot grabbed algae, threw it aside, then placed coral. The claw cracked during a Central Missouri match; the team fitted a spare, and the event machine shop made another plate. Motors and ratios unconfirmed (see CAD).	**INTAKE** Coral intake method unconfirmed (see CAD). Before the Championship the team added a ground algae intake to Dobson.
ELEVATOR / ARM Elevator. At Central Missouri the team tensioned the elevator too tight and it could not extend fully. For that match it ran an L2 auto and scored mostly on L3; the elevator was then re-tensioned and re-tuned. In off-season autos the code raises the elevator before the robot reaches the reef. Stages and motors unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). At Greater Kansas City the team had gyro drift; it used workarounds while the cause was found between matches, with help from the CSAs and CTRE.
ENDGAME Climb type unconfirmed (see CAD). The team states it improved its endgame for Greater Kansas City and climbed 3 times there.	**AUTONOMOUS** At Central Missouri the robot ran a 1-coral auto, with an L2 auto as a fallback; it first completed its 2-coral auto in Qual 33. The team states its scouting data showed the most consistent auto at Greater Kansas City. More center autos were added before the Championship. Off-season code reached a 3.5-coral auto, built with a string-driven auto builder from preset field positions and some on-the-fly pathfinding.
VISION A camera on a TPU mount feeds vision updates to odometry for alignment. At Central Missouri the sponsor plate flexed down and bent the mount, and vision put the robot a few inches off. In a later match the camera was misaligned again and the driver scored manually. Camera type unconfirmed.	**PUBLIC CAD / DESIGN** Fusion 360 CAD

LESSONS FOR STUDENTS

- Keep body panels away from sensor mounts: a flexing sponsor plate bent the TPU camera mount, and vision then moved odometry a few inches off.
- Record elevator tension settings: an over-tensioned elevator stopped short of full height, and the team fell back to an L2 auto.
- Bring spares of end effector plates: the algae claw cracked mid-match, and the team swapped in a spare from the pit.
- A string-driven auto builder ('A4P0B4') turns preset positions into one sequential command; the team states it removed most path-planning time.
- Change the robot to fit the alliance: before the Championship the team added a ground algae intake and more center autos.

Fusion 360 CAD: Fusion 360 CAD · Fusion 360 CAD · 2025 build thread (CD) · Code (GitHub) · Teaser thread (CD) · 2026 build thread with the 2025 code post (CD)

■ Study pick

3357 COMETS Tempel

USA-MI

Local EPA 56 (#490)

23-21

'Tempel': 24x24 in frame, ground coral, sideways belt-in-tube elevator, wrist with coaxial roller drive, deep climb

SCORING (CORAL / ALGAE) Coral rollers on a wrist. An outer shaft drives wrist position through a 20t/80t 32DP gear set. A shaft inside the 2x1 tube drives the rollers through a bevel gearset, then 20DP gears, then a belt and pulley set. The wrist rotates about the roller shaft; the OP compares the layout to a differential.	**INTAKE** Ground coral pickup, stated in the reveal post. One CAD screenshot in the release post is titled 'Drivebase-AlgIntake'. Intake motors and geometry unconfirmed (see CAD).
ELEVATOR / ARM Sideways-mounted elevator with a continuous belt-in-tube drive. UHMW slider pucks and clamping plates are similar to the team's 2022 and 2023 designs. The team replicated the gearbox and idler layout from a published COTS-compatible belt-in-tube elevator design and states it is easy to rig and tension.	**DRIVETRAIN** 24 in x 24 in frame perimeter. Bumpers have chamfered corners, meant to pivot and roll off defenders, and slide latches. Drivetrain type unconfirmed (see CAD).
ENDGAME Deep cage climb with a deployable foot and arm. Surgical tubing pulls a lever arm, and a ratchet on the shaft holds the foot out. The team reported it as mostly functional but fussy to retract after the match, and named a torsion spring as a possible fix.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** In-season CAD (GrabCAD)

LESSONS FOR STUDENTS

- Drive a wrist and its rollers through coaxial shafts: wrist angle from a 20t/80t 32DP gear set on the outer shaft, rollers from a bevel gearset on an inner shaft inside the 2x1 tube.
- Reuse a published elevator design: the team replicated the gearbox and idler layout of a COTS-compatible continuous belt-in-tube elevator and kept its own UHMW slider pucks.
- A climb foot can be deployed by surgical tubing on a lever arm and held out by a ratchet. The team reported it as fussy to retract after the match.
- Small frame: a 24x24 in frame perimeter, with chamfered bumper corners meant to roll off defenders.

GrabCAD: In-season CAD (GrabCAD) · Reveal and CAD release thread (CD) · In-season CAD (GrabCAD) · Reveal video

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Profile

2485 Overclocked Nessie

USA-CA

Local EPA 53 (#581)

20-16

'Nessie': elevator robot with a bent polycarbonate coral ramp into the end effector and a cage climber

SCORING (CORAL / ALGAE) Coral end effector fed by a bent polycarbonate ramp. Algae handling: Unconfirmed (see CAD).	**INTAKE** A bent polycarbonate ramp, added after Week 4, funnels coral into the end effector. Ground intake: Unconfirmed (see CAD).
ELEVATOR / ARM Elevator. After Week 4 the team ran one motor for elevator extension; the team states this is why the CAD shows interference between one elevator motor and the climber motor.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Cage climber. After Week 4 the team changed the climber geometry to guide the cage around the subsystem. Deep or shallow cage: Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Track in-season revisions: the release lists three changes after Week 4 so a reader can compare the magazine with the CAD.
- A bent polycarbonate ramp can guide coral into an end effector; the team added one after Week 4.
- The team switched to Onshape in 2025 to increase younger student involvement, because collaboration is easier.

Onshape CAD: Onshape CAD · Onshape CAD · 2025 Robot Magazine, Week 4 (Google Drive)

■ Study pick

3669 CarbonKnights Twitchy

USA-CA

Local EPA 52 (#593)

9-13-1

'Twitchy': two parallel carbon fiber tube elevators with 3D printed bearingless sliders; built without CNC

SCORING (CORAL / ALGAE) In offseason testing the team added a command that scores at the closest left or right reef pole using one Limelight. End effector type and motors: Unconfirmed (see CAD).	**INTAKE** Unconfirmed (see CAD).
ELEVATOR / ARM Multistage elevator built from carbon fiber tubes and 3D printed parts, inspired by HighTide's 2023 elevator. The plastic parts slide directly on the tubes with no bearings. The team states linear bearings on a multistage would cost too much weight. Two elevators run in parallel on opposite sides of the robot.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The team made drivetrain improvements before its San Diego offseason event; it does not describe them.
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Single Limelight using MegaTag2 pose data, made to work over the 2025-26 Christmas break. A 2023 attempt with MegaTag1 failed because the pose data was too noisy. The current code is on the LimelightTesting branch.	**PUBLIC CAD / DESIGN** 'Twitchy' robot CAD (Onshape)

LESSONS FOR STUDENTS

- A multistage elevator without CNC: 3D printed parts slide on carbon fiber tubes with no bearings.
- Count the cost of parts: the team states linear bearings on every stage would cancel the weight savings, so it dropped them.
- Two elevators in parallel on opposite sides of the robot add strength and stability.
- Pose estimation changes between versions: MegaTag1 was too noisy in 2023; MegaTag2 worked for automatic reef scoring.
- Driver input is part of the design: the team built a controller from an Arduino and a RadioMaster radio's gimbals for finer low-speed control.

Onshape CAD: 'Twitchy' robot CAD (Onshape) · CAD and code release (CD) · 'Twitchy' robot CAD (Onshape) · Standalone elevator CAD (Onshape) · Code (GitHub) · Custom controller code (GitHub)

Profile

1086 Blue Cheese Stringray

USA-VA

Local EPA 50 (#657)

31-25-1

'Stringray': coral station intake, coral L1-L4, algae from reef to net and processor, deep cage climb

SCORING (CORAL / ALGAE) Scores coral on all 4 reef levels. Scores algae in the net and in the processor. End effector type, motors and ratios: Unconfirmed (see CAD).	**INTAKE** Coral intake from the Coral Station. Removes algae from the reef. Ground intake: Unconfirmed (not stated in the public release).
ELEVATOR / ARM Unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Climbs on the deep cage. Climber mechanism: Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Unconfirmed (not stated in the public release; see code).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One robot covers every task in the release list: Coral Station intake, L1-L4 coral, algae removal, net and processor scoring, deep climb.
- CAD and code are both public, so each listed capability can be traced to its mechanism and its code.
- The release names capabilities, not mechanisms; the superstructure and end effector must be read from the CAD.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub)

Profile	1740 Cyber Colonels USA-CT	Local EPA 42 (#988) 12-17-1

January plan: coral-only RustHOUNDS Ri3D-style robot, passive ramp intake, elevator with pivoting dunk arm; WCP Swerve X2

SCORING (CORAL / ALGAE) Passive end effector on a pivoting arm that 'dunks' coral onto the reef, after the RustHOUNDS Ri3D concept. The team wanted to look at letting the coral staging shelf also eject coral onto L1, and considered a separate algae removal arm on the fixed elevator stage. Plan as of 2025-01-12; check the CAD for the final version.	**INTAKE** Planned as of 2025-01-12: passive ramp intake at the coral station. 'Just chuck it in' human player loading was a kickoff need. Ground pickup was listed as nice to have.
ELEVATOR / ARM Elevator with a pivoting arm. The team tested dunk behavior on a lumber, cart and 3D printed test rig, trimming a PVC arm to compare lengths. Elevator kit make unconfirmed (see CAD).	**DRIVETRAIN** WCP Swerve X2 modules, assembled by students, with Kraken motors (model not stated). First built as a 30 in x 30 in chassis with cross rails for the elevator; the team expected it might be cut down by 2 in.
ENDGAME Shallow cage climb was a need and deep climb a want at kickoff. By 2025-03-18 the team was redesigning its climber. A mentor post shows a prototype: 3D printed plates with engineered flexures and stealth wheels that pull the cage bar in under motor torque. With motor braking or a ratchet, the plate stiffness holds the bar like a latch. The part is intended to screw on in place of an Everybot harpoon hook and be winched in. An earlier cutting-board version lifted the robot, then failed at a thin section left by a CNC router setup error. Whether this climber competed: Unconfirmed (see CAD).	**AUTONOMOUS** Targets stated at kickoff: an auto that leaves and scores L1, and one that scores L1 and attempts L4. Autonomous scoring and loading alignment was a need. Final routines unconfirmed.
VISION Orange Pi and cameras bought for PhotonVision testing; on 2025-01-10 students were mounting them on the 2024 robot for reef and loading station alignment work. Camera count unconfirmed. The software lead chose AdvantageKit to review and replay logs because full-scale testing time was short.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Derive the cycle target from match time: about 105 s over 8 coral gives about 12 s per cycle, with 6 s of that for travel at max speed.
- Limit scope to protect testing time: a 2024 add-on cost testing time and reliability, so the team chose to focus the robot on one aspect. It dropped algae scoring because it expected many algae robots.
- Prototype motion cheaply: a lumber, cart and 3D printed rig with a PVC arm trimmed inch by inch let the team compare dunk arm lengths before CAD.
- Reinforce printed test fixtures: reef branches with 3/4 in steel conduit inside the PVC upright were stiff enough to use; PVC alone was too flexible.
- Compliant parts can latch: the mentor states printed flexure plates with stealth wheels hold the cage bar like a latch when motor braking or a ratchet stops the wheels. An early cutting-board version failed at a thin section caused by a router setup error, so check toolpaths on flexure parts.

Onshape CAD: Onshape CAD · Onshape CAD · Build thread (CD) · Code (GitHub) · Team YouTube channel

■ Study pick

614 Night Hawks Stand Mixer

USA-VA

Local EPA 41 (#1010)

19-24-1

'Stand Mixer': 2-in-1 ground coral intake (L1 hold or pass-through) feeding a 'Canal' to the end effector

SCORING (CORAL / ALGAE) The end effector receives coral from the Canal. It detects a held coral when its motor shows a current spike and the LaserCAN no longer sees the coral. Algae handling: Unconfirmed (see CAD).	**INTAKE** Ground coral intake with two modes. A 0.85 in OD, 0.65 in ID PVC pipe stops the coral, so the driver can score it on L1 from the intake. A button spins the rollers faster; the PVC flexes and the coral passes through to the Canal. The Canal also takes coral from the Coral Station when the human player drops it in.
ELEVATOR / ARM The Canal carries coral from the intake to the end effector; a LaserCAN on the belly panel trips as the coral passes. A non-team reply describes a single-stage elevator; the team did not confirm it. Elevator or arm detail: Unconfirmed (see CAD).	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release). Coral sensing uses a LaserCAN and motor current.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- One intake can serve two jobs: the PVC stop holds coral for L1, and a higher roller speed flexes it to pass coral through.
- Combine signals for state detection: the end effector uses a motor current spike plus a LaserCAN on the belly panel to confirm a held coral.
- The team names its own gap: no sensor in the intake. It would add one, and more control, to stop unintended pass-throughs.
- One path for two sources: the Canal takes coral from both the ground intake and the Coral Station.

Onshape CAD: Onshape CAD · CAD and binder release thread (CD) · Onshape CAD · CAD and tech binder (team website)

Profile	7312 T3 Nessie USA-TX	Local EPA 35 (#1349) 24-18

'Nessie': 3-stage GreyT cascade elevator; elbow and wrist pivots; belted coral/algae end effector; swerve

SCORING (CORAL / ALGAE) End effector of compliance wheels linked by belts that holds both coral and algae. The first version was too large, so the team revisited a design similar to Spectrum's intake and turned it 90 degrees. In testing it took coral from the source in both orientations and held algae when shaken. It mounts on a MAX spline shaft at the wrist pivot; motors unconfirmed (see CAD).	**INTAKE** Coral comes from the source into the end effector; the early prototype also picked algae from the ground and the reef. A slap-down ground coral intake concept was dropped for complexity, coral jam risk and poor fit with autonomous tasks.
ELEVATOR / ARM Three-stage GreyT Cascading Elevator from WestCoast. An elbow pivot on the carriage turns the arm on a MAX spline shaft through 2:1 sprockets, belt-driven by a REV MAX 90-degree gearbox, with an absolute encoder. A wrist pivot uses a 90-degree gearbox at 2:1.	**DRIVETRAIN** Swerve. The programming team coded a test swerve base and reported it ready to go on the competition robot. Module make and ratio: unconfirmed (see CAD).
ENDGAME Cage climber inspired by the RustHOUNDS Ri3D climber: a pivot with a metal pole, turned by a 90-degree gearbox at 200:1. The pole enters the cage, rotates the cage and keeps turning until the robot is off the ground. Deep or shallow cage: unconfirmed; the team's priority list ranked the shallow cage above the deep cage.	**AUTONOMOUS** Unconfirmed (not stated in the public release)
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write down why concepts lose: the team dropped a slap-down intake with a telescoping arm for complexity, coral jam risk and poor fit with autonomous tasks.
- Reorient before redesigning: the first end effector was too large, and turning a Spectrum-like design 90 degrees fixed that design's coral scoring and arm mounting problems. The result takes coral in both source orientations.
- Rank scoring tasks first: the team's priority list put L2-L4 and algae removal above L1, processor, deep cage and net.
- Two pivots on a COTS elevator: a three-stage GreyT cascade carries elbow and wrist pivots, each at 2:1 off a 90-degree gearbox.

Onshape CAD: Onshape CAD · Onshape CAD · Reveal video 'Nessie'

Profile	2079 4H ALARM Robotics USA-MA	Local EPA 29 (#1774) 9-19-1

28x28 swerve, 2-stage cascade elevator, wrist carriage with top-down roller end effector, rear coral funnel, deep cage climber

SCORING (CORAL / ALGAE) Carriage with a 'goose neck' plate and a wrist, the design the team was working on in Week 3 after a swing-through arm found no workable fixed angle in CAD. Changed after the V1 build (post of 2025-02-22): the V1 end effector had trouble intaking coral, removed algae slowly and took longer to score L4. The team remodeled the end effector as V2 with a top-down roller arrangement. The team states V2 removes algae more easily and gives better grip and coral centering in most scenarios.	**INTAKE** Coral from the human player station; in Week 3 the team leaned toward the view that ground coral pickup might not be worth the added weight. V2 (2025-02-22) added a passive funnel at the back of the robot for human player coral drops and for intaking under defense. Algae ground intake on a pivot, as of 2025-02-03: two side plates (0.22 in polycarbonate or 0.125 in aluminum) with a roller, and a hex standoff at the bottom in place of a second roller.
ELEVATOR / ARM 2-stage cascade elevator, chosen over a telescopic arm because telescopic arms had not worked for the team in past years. The carriage carries the goose-neck plate and wrist.	**DRIVETRAIN** 28 x 28 in swerve base, sized for stability when extended to score L4. The team added 1x1 standoffs to lower the base plate from about 1.2 in to about 0.2 in off the floor, after 4481 reported robots lifting on algae. The team states this also lowered the COG. Module type unconfirmed (see CAD).
ENDGAME Deep cage climber with a low hinge. The arm pivots at the bottom, is pulled down by a string and lifted by a spring. The cage clamp moved from one pivot to two pivots to reduce the force needed to slide onto the cage. The 1x1 tube arm weighs 1.28 lbs.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Planned in Week 3: PhotonVision on 4 Arducams, one per side, for full-field localization, with Limelights for game-piece tracking. Final setup unconfirmed.	**PUBLIC CAD / DESIGN** 2025 CAD (Onshape)

LESSONS FOR STUDENTS

- Sort needs from wants first: swerve, all four reef levels and a climb were needs; ground coral, deep cage and a wrist were wants.
- Model the whole robot against the reef and the station before committing. The team dropped its fixed-angle arm when no angle worked in CAD.
- Two pivots on a spring-loaded cage clamp reduced the force needed to slide onto the cage.
- Track weight per part: the 1x1 tube climber arm is 1.28 lbs, about 5 lbs less than the 2x1 version.
- Test bumper materials on a repeatable rig: a cage hit a 30 lb box; EVA foam and polyethylene held up better, and a combination of both held up better still.
- Build V1 early and judge it on the field tasks: the V1 end effector struggled with coral intake, algae removal and L4 time, which led to the top-down roller V2.

Onshape CAD: 2025 CAD (Onshape) · 2025 CAD (Onshape) · 2025 build thread (CD) · Code (GitHub) · Climber prototype run (video) · Bumper material test (video)

Profile	7691 TSA All Stars USA-TX	Local EPA 25 (#2124) 18-24-1

Swerve; elevator with arm end effector for coral; pivoting ground coral intake; side algae grabber; deep climber (later removed)

SCORING (CORAL / ALGAE) An arm on the elevator sets the coral angle for each reef level. The end effector and its backup broke in the third match at Houston. A rebuilt end effector used a stronger 3D printing filament and also failed. Motors and ratios unconfirmed (see CAD).	**INTAKE** Over-the-bumper ground coral intake on a pivot, called the 'dustpan', which hands coral to the elevator. A separate attachment on the side picks up algae and carries it to the processor. The team states the intake and handoff did not work as designed. At Space City the intake broke and the team played two matches as defense; coral and algae falling into the robot cost the team multiple matches there.
ELEVATOR / ARM Elevator, the team's first. The team states the elevator was not braced enough. At Houston they planned 1x1 MAXTube bracing with 3D printed gussets as a quick fix. Stages and motors unconfirmed (see CAD).	**DRIVETRAIN** Swerve: the week 4 post describes base code with vision and swerve subsystems. Module type unconfirmed (see CAD). A swerve motor broke at the offseason TRI event.
ENDGAME A deep cage climb mechanism at the rear, with a design guided by 118's Everybot. Before the District Championship the team worked on the code and the climb, but states it did not work. The team later removed the climber and the baseplate because the robot was overweight.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION The coding team worked on AprilTags for the first time; earlier seasons relied mostly on driver sight and a camera when it worked. Camera and coprocessor unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Design the connections, not only the parts: the team states its subsystem CAD was sound but the integration, wiring and handoff were not.
- Weight budget early: the robot was overweight and the team removed the baseplate and the climber.
- Brace a first elevator in CAD: the team found its elevator under-braced at its first event.
- Back up CAD: the team reports its Onshape file was destroyed during the Houston event, which forced a re-CAD in the pit.
- Close openings where game pieces can fall in, and secure the battery: falling coral and algae cost the team matches at Space City, and unplugged batteries cost matches at the District Championship.
- The season review led to new rules: the CAD subteam now requires a year of robotics experience to join, and the team plans to build only from CAD checked by an audit system.

Onshape CAD: Onshape CAD · Onshape CAD · Onshape document linked in the 2025 build thread · 2025 build thread (CD) · 2025-2026 build thread with the season review (CD)

■ Study pick

1806 S.W.A.T. Atom

USA-MO

Local EPA 10 (#3554)

9-16

'Atom': swerve, custom welded elevator; rebuilt mid-season toward a 2056-style coral/algae claw on an arm

SCORING (CORAL / ALGAE)

Week 2 design: a 3-DOF coral claw with an intake motor, an open/close motor, and rotation from inside to outside the robot. The first prototype used 8 in compliant wheels with no grip on coral. 3D printed gears broke in version 4; in Week 5 the team redesigned both claws without printed gears. Changed after Central Missouri: the plan became one claw for algae and coral on an arm, inspired by 2056. The team assembled a '2056 style claw' by 2025-03-13 but says the rebuild was not finished for Greater Kansas City. The retrospective says the arm pivot was too floppy and had too much backlash for fast or precise motion.

INTAKE

The Week 2 concept had a ground coral intake on the right side and a ground algae intake at the front, with coral station loading kept. The concept changed in Week 3; the team lists slightly narrower ground intakes as a con. Changed after Central Missouri: a floor coral intake inspired by 1690 and 118, plus a coral righting mechanism tested on 2025-03-12. At Greater Kansas City the intake chain bound; the team says the double sprockets did not seem to have enough spacing. The team removed the intake and used a feeder station funnel.

ELEVATOR / ARM

The Week 2 concept put the elevator at the back of the robot. Elevator based on the team's 2018/2019 design: 2x1x1/8 in C-channel stitch welded to 1.5 x 1/8 in flat bar, with delrin/acetal rollers on shoulder bolts. In Week 5 the team moved to side rigging with 4 spools and cascade rigging. The elevator calculation linked in the thread uses 2 Kraken X60 at a 5.333 reduction. The rebuild plan kept this elevator with fewer stages, because the arm reaches higher. At Greater Kansas City, 3/16 Dyneema could not lift the elevator; team 1802 lent 1/16 steel aircraft cable, and the V-groove bearings held it. Cable tension from the rigging angles stopped the elevator from going down much below its starting height.

DRIVETRAIN

28 x 28 in frame. Swerve, per the team's posts: YAGSL configuration and 'snap angles' heading presets at 60 deg steps for the reef. The practice bot tested 3D printed 95A TPU wheels. On the competition robot the 3D printed wheels were slicks by the 2nd match of defense at Central Missouri. Module type unconfirmed (see CAD).

ENDGAME

The Week 2 plan was a cage-shaped hole in the belly pan and a claw on one elevator stage. By Week 3 the team had a prototype climber claw that uses the elevator to lift. The climber hook was in CAD by Week 5. Climb at events, and deep or shallow cage, unconfirmed.

AUTONOMOUS

In Week 5 the team tested paths in AdvantageKit simulation for a 4-coral auto, all from the ground, ending with an algae run to the processor. The team calls these paths 'probably wishful thinking'. The retrospective says controls students had almost no time with the robot. Competition autos unconfirmed.

VISION

The team mentions a vision PC. Its Ethernet port burned out 2 days before Greater Kansas City; the team advises turning off PoE on the radio when it is not used. Cameras and software unconfirmed.

PUBLIC CAD / DESIGN

'Atom' CAD (Onshape)

LESSONS FOR STUDENTS

- Check rotation direction before printing parts: the first claw prototype had its wheels spinning in opposite directions.
- Test printed parts under load early. Printed claw gears broke under drill testing, and printed wheels were slicks by the 2nd match of defense.
- Simulation needs the current configuration: the robot did not move in YAGSL simulation until the team entered this year's gear ratios.
- Size the elevator cable to the load: 3/16 Dyneema could not lift the elevator, and 1/16 steel aircraft cable did.
- The thread lists pros and cons for the Week 3 concept change. The retrospective says the team over-estimated its resources and needs to fail faster and count DOFs to judge complexity.

Onshape CAD: 'Atom' CAD (Onshape) · 'Atom' CAD (Onshape) · Code (GitHub) · 2025 build thread (CD) · Week 5 update video · Intake testing video (Instagram)

Sources: the Spectrum 3847 CAD Collection, team CAD, binder and code releases, Chief Delphi release and reveal threads, and match data from The Blue Alliance. EPA is Local EPA (approx), computed independently. It is not official Statbotics data. "Unconfirmed" means the public material does not state the detail.